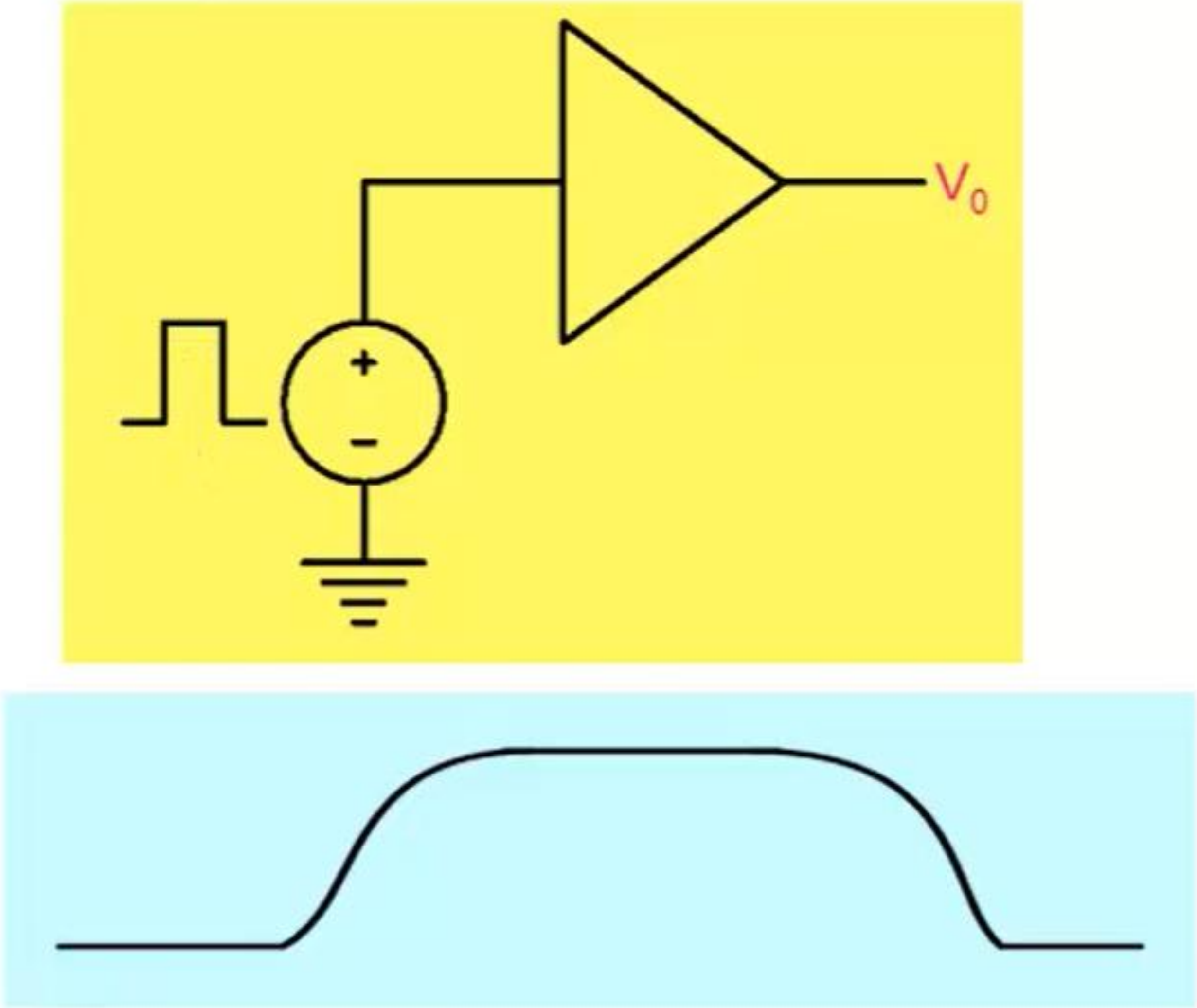
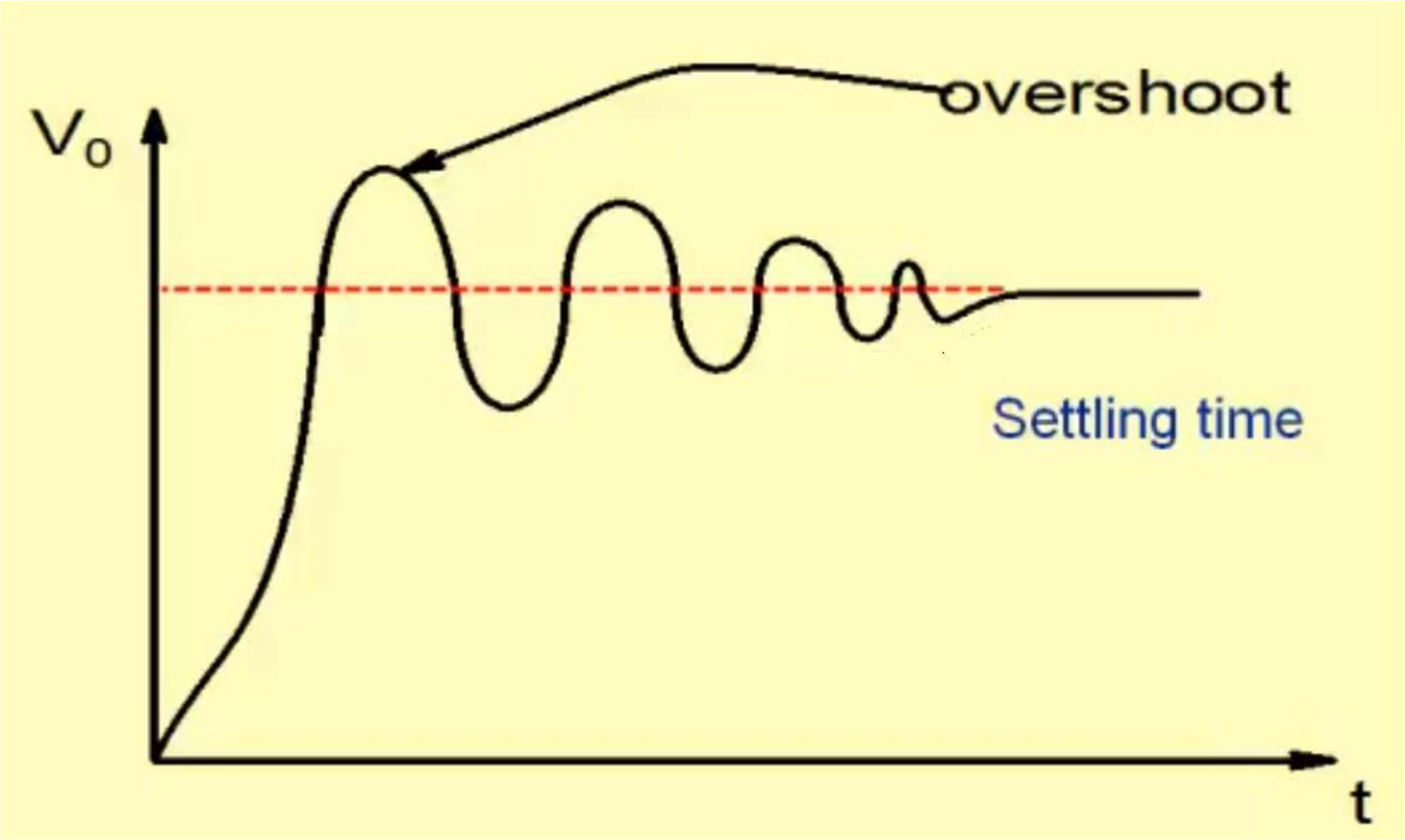
Transient Response



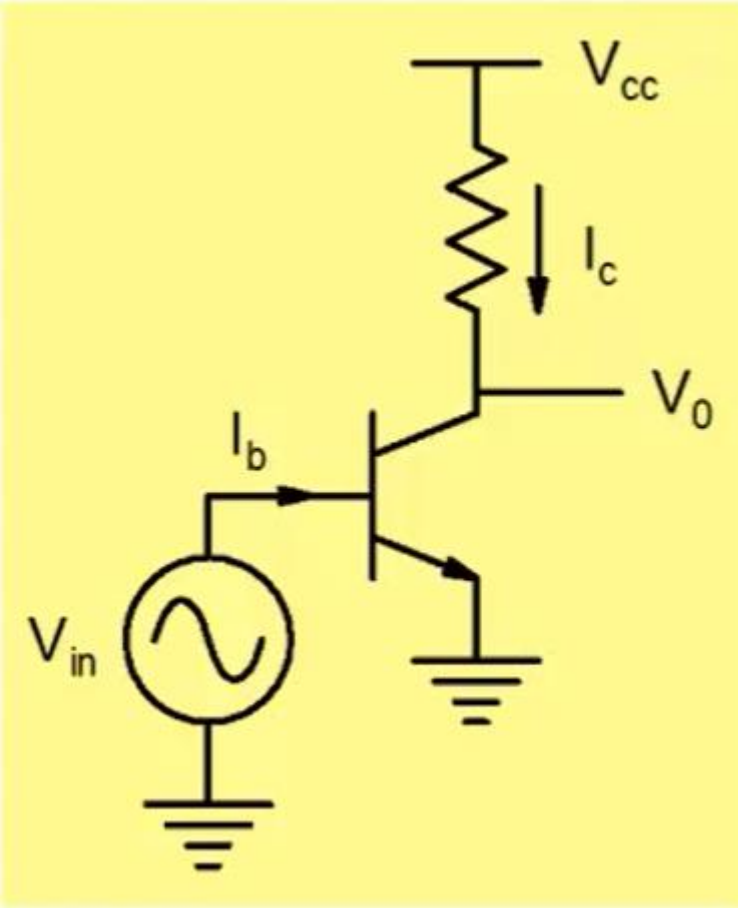
Rise time and fall time

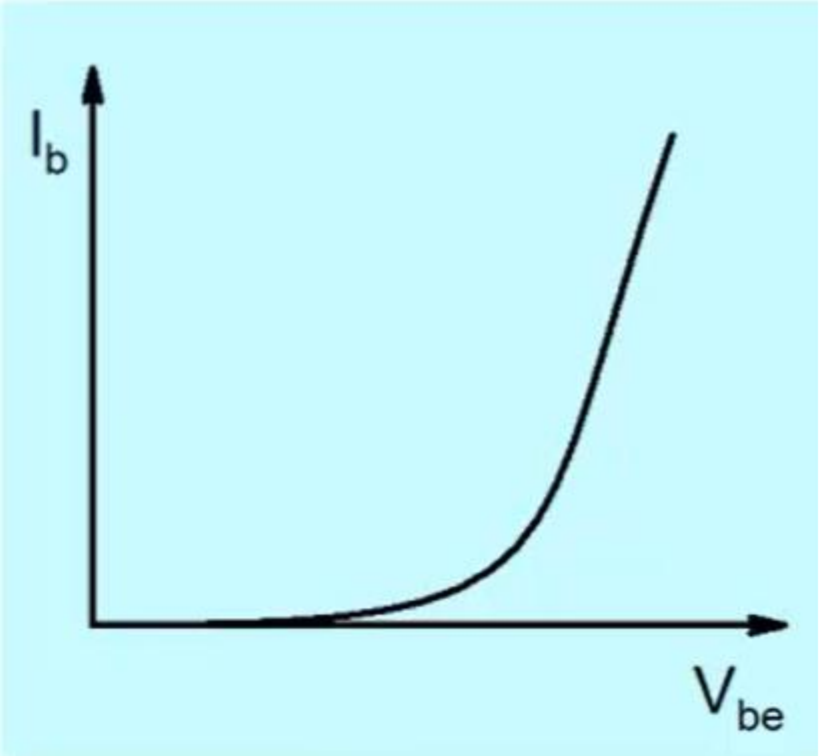
Overshoot and time Settling time

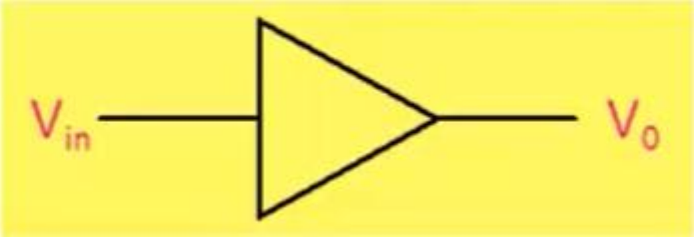


Distortion:

All amplifiers are nonlinear because transistors used for building amplifier are nonlinear elements.







$V_{in} = a_0 \sin wt$

$V_0 = b_0 + b_1 \sin \omega t + b_2 \sin 2\omega t + b_3 \sin 3\omega t + \ldots\ldots\ldots$

$$HD_2 = \frac{b_2}{b_1} \times 100$$

$$HD_3 = \frac{b_3}{b_1} \times 100$$

$$THD = \frac{\sqrt{b_2{}^2 + b_3{}^2 + \ldots\ldots}}{b_1} \times 100$$

Example

$$V_0 = kV_{in} + \frac{k}{10} V_{in}^{\;2}$$

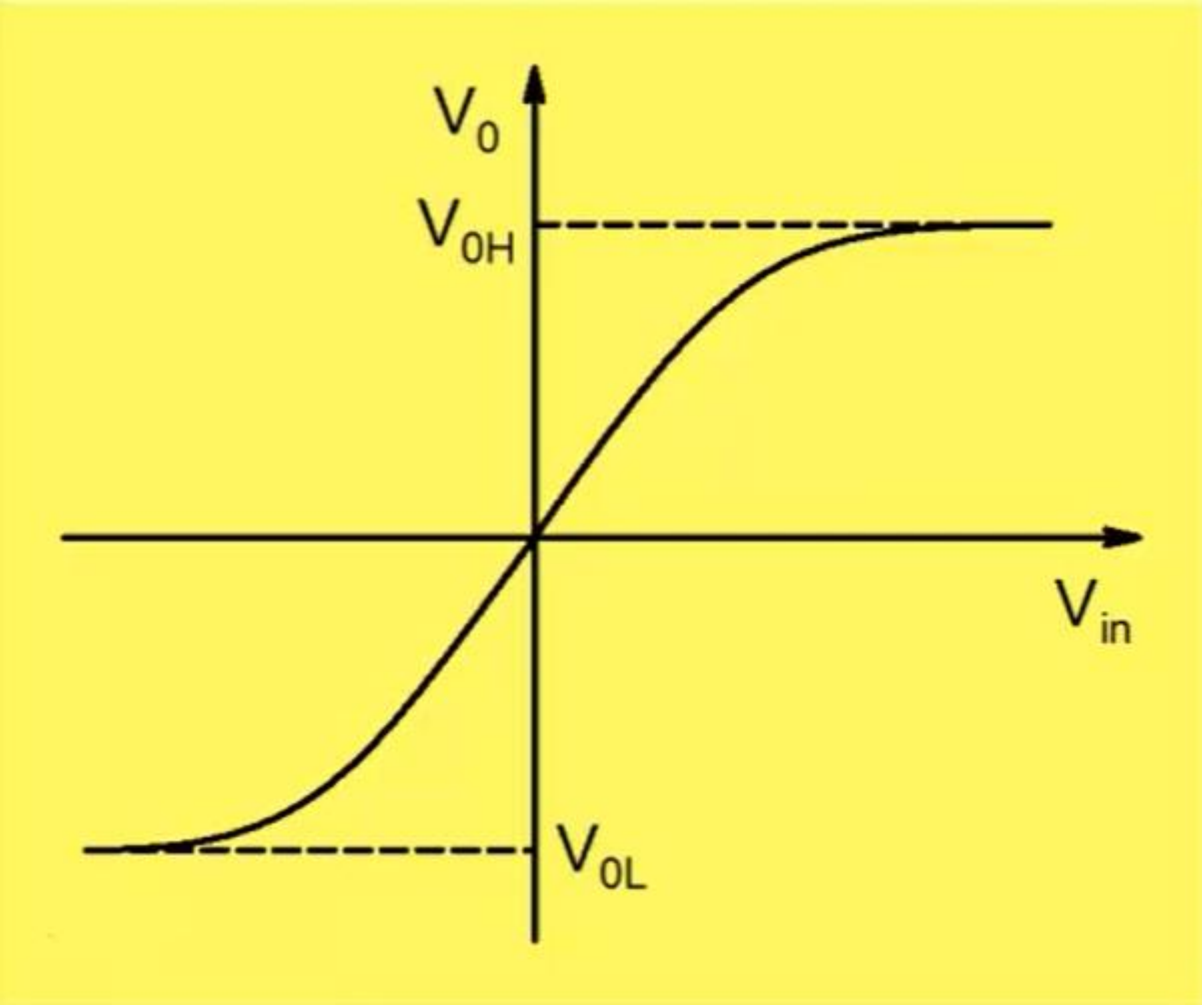
$$V_{in} = a_0 \sin \omega t$$

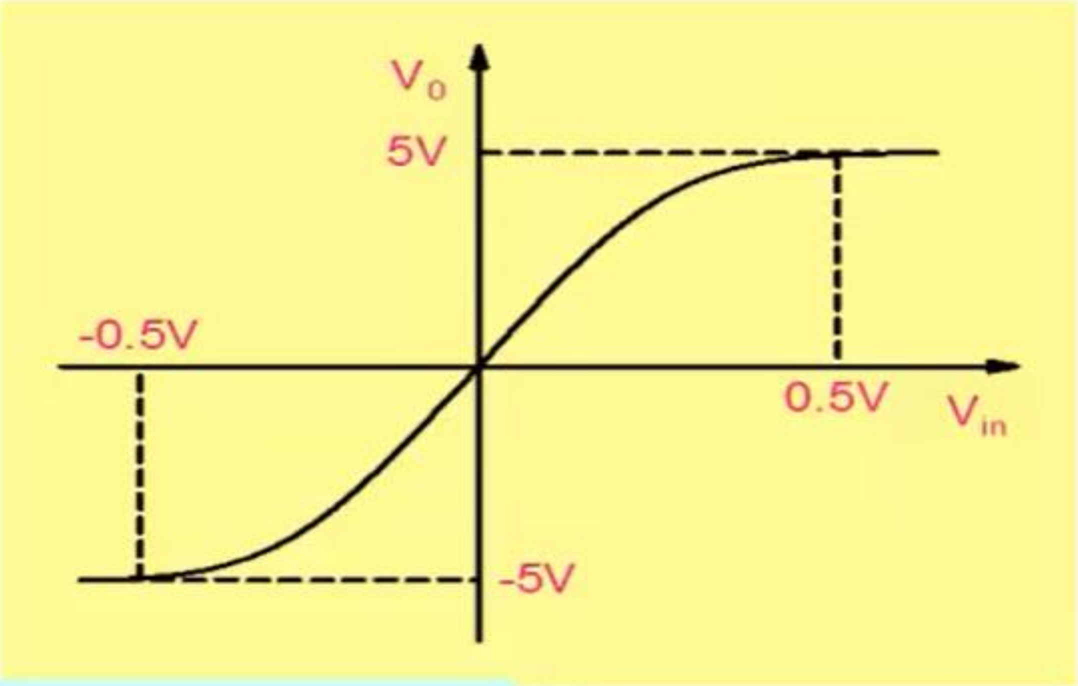
$$V_0 = \frac{ka_0^{\;2}}{20} + ka_0 \sin \omega t - \frac{ka_0^{\;2}}{20} \cos 2\omega t$$

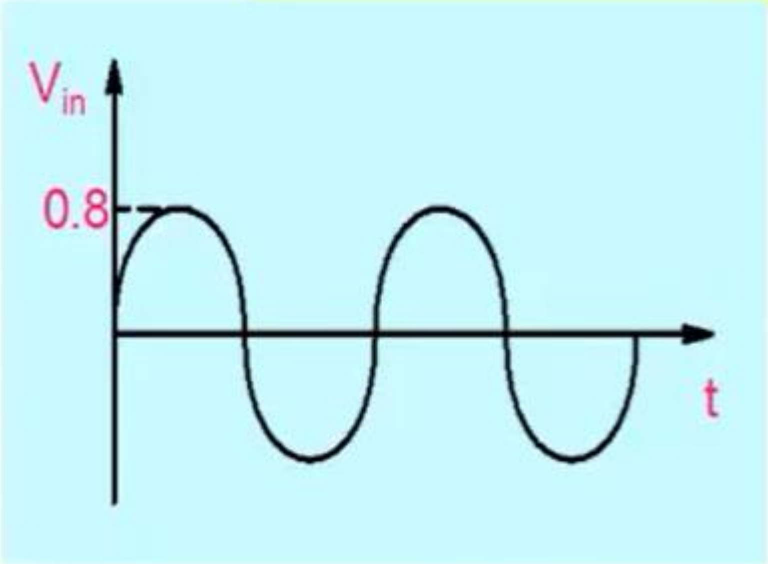
$$THD = HD_2 = \frac{ka_0^{\;2}/20}{ka_0} \times 100 = 5a_0$$

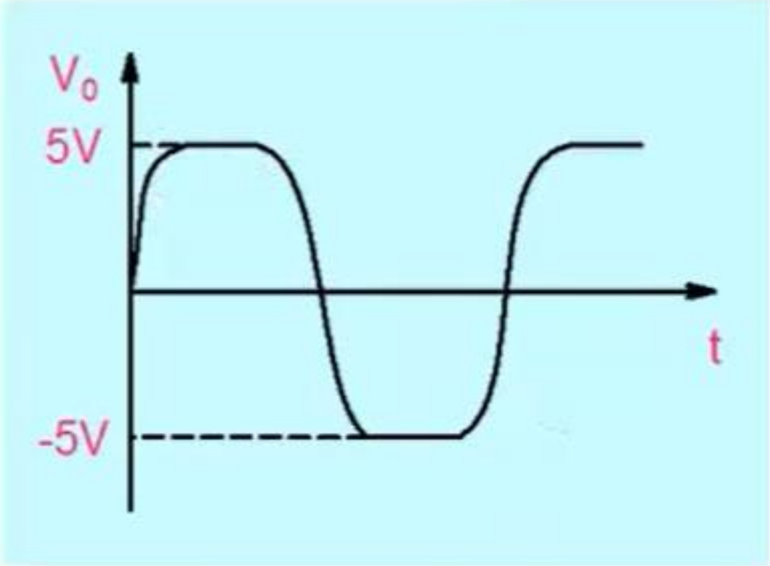
Distortion increases with magnitude of input signal !

Maximum Voltage Swing



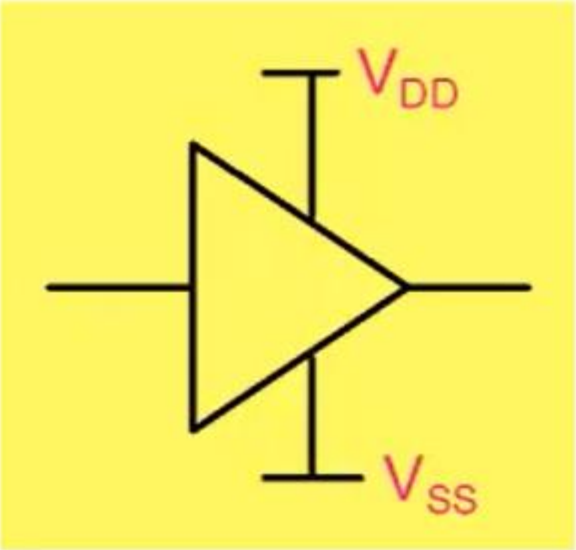

V_o
5V
-0.5V
0.5V
V_{in}
-5V

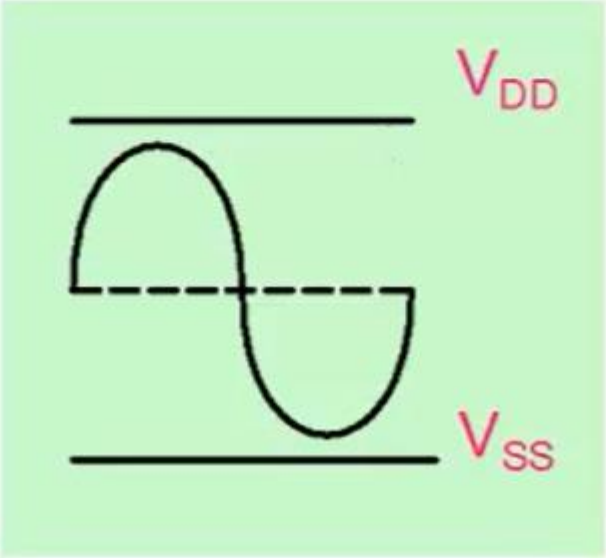

V_{in}
0.8
t


V_o
5V
t
-5V

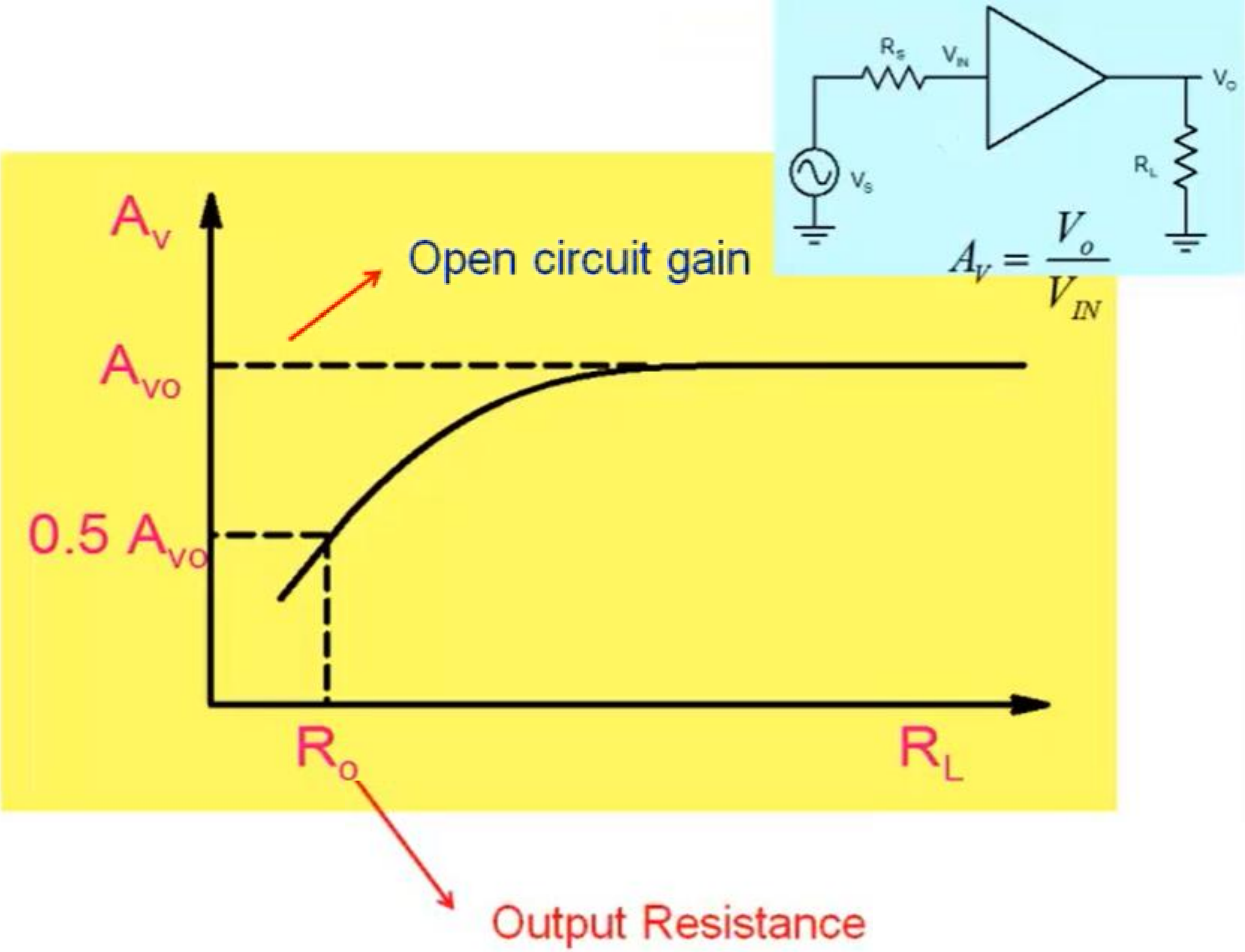
Rail-to-Rail output voltage swing

$V_{OH} \leq V_{DD}$ $V_{OL} \geq V_{SS}$

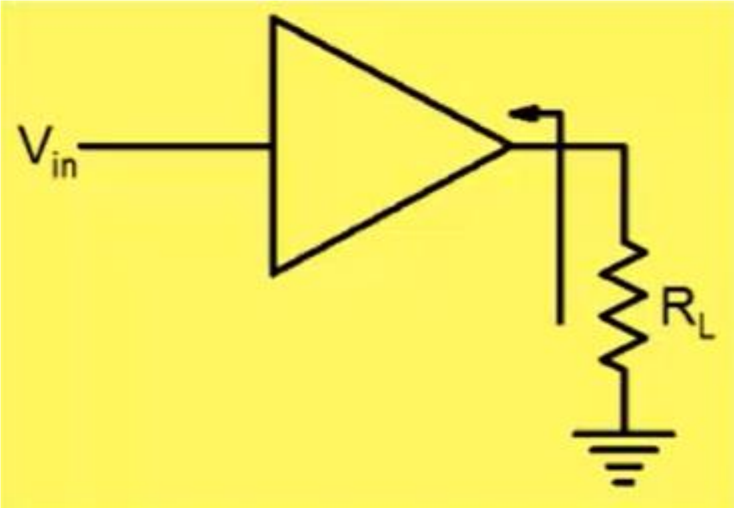


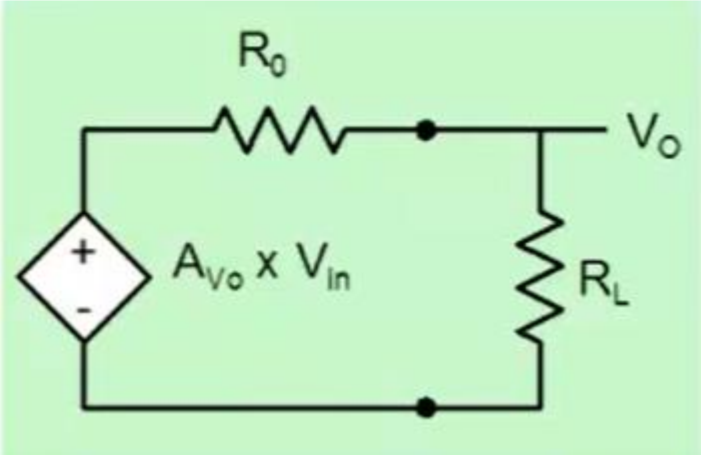


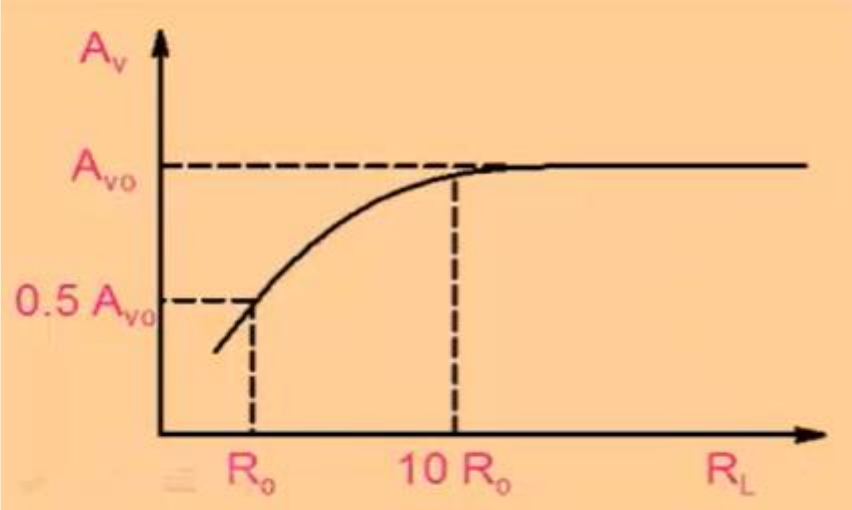
Effect of Load Resistance



Open circuit voltage gain and output resistance

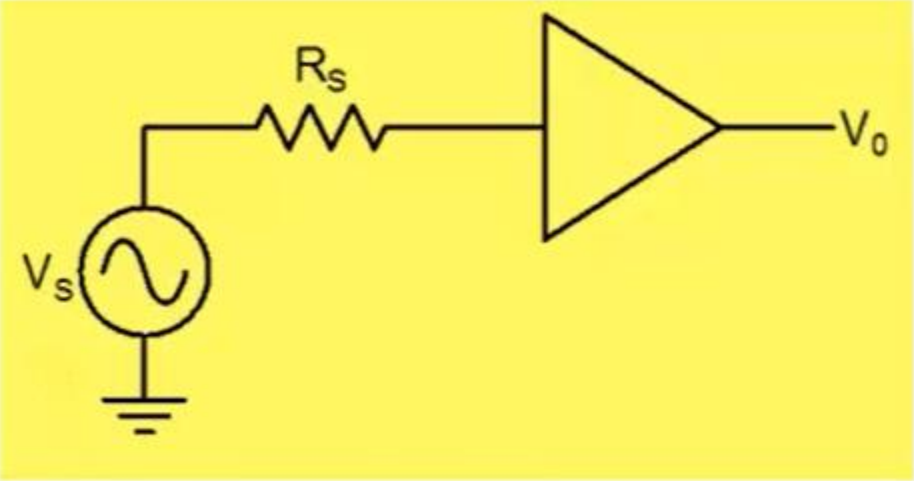




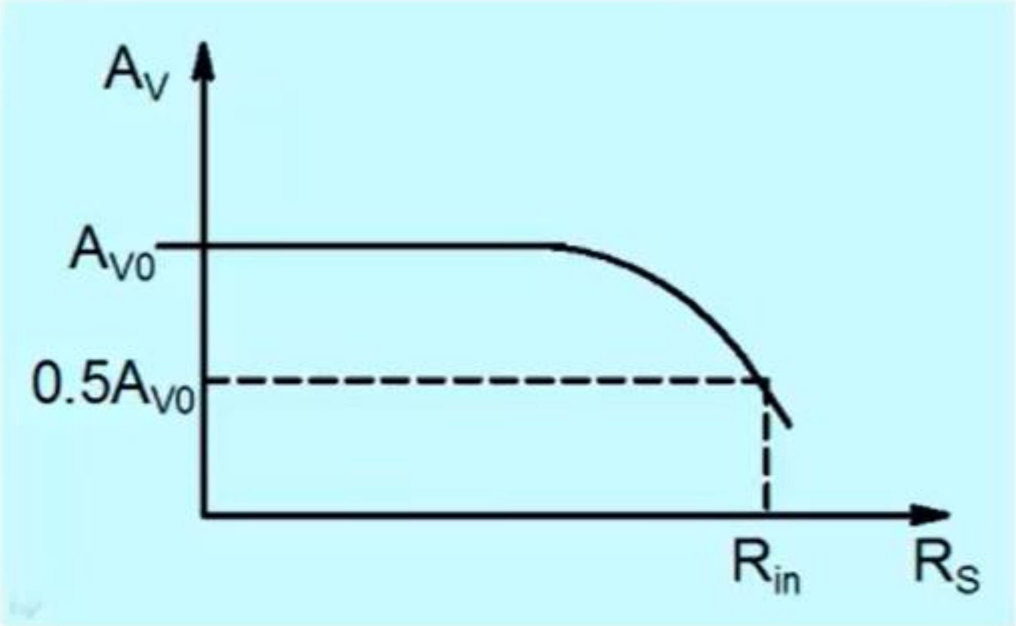


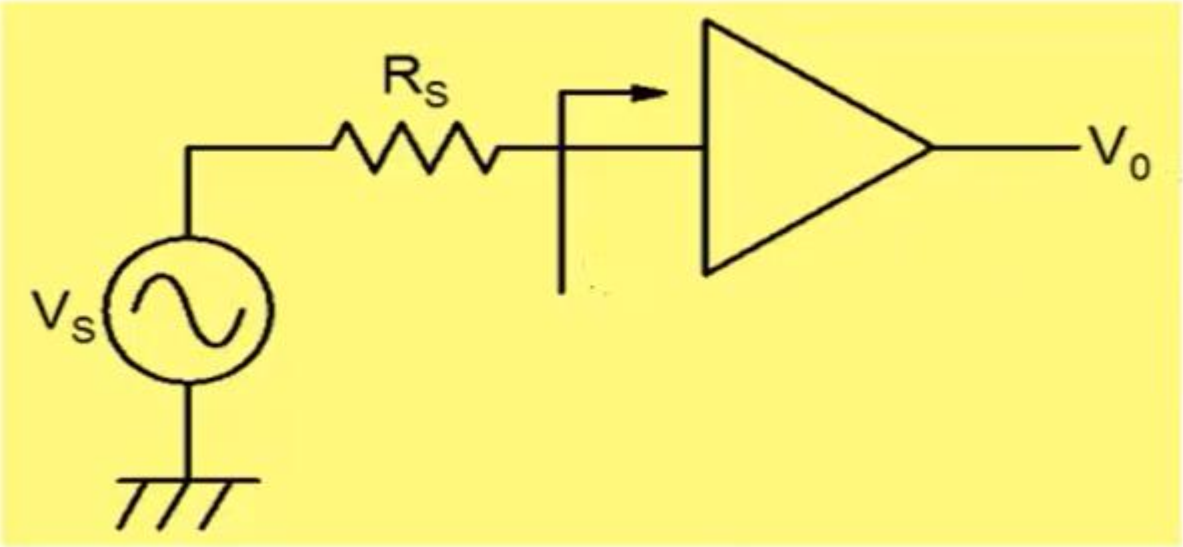
$$V_0 = A_{V0} \times V_{in}\left(\frac{R_L}{R_0 + R_L}\right)$$

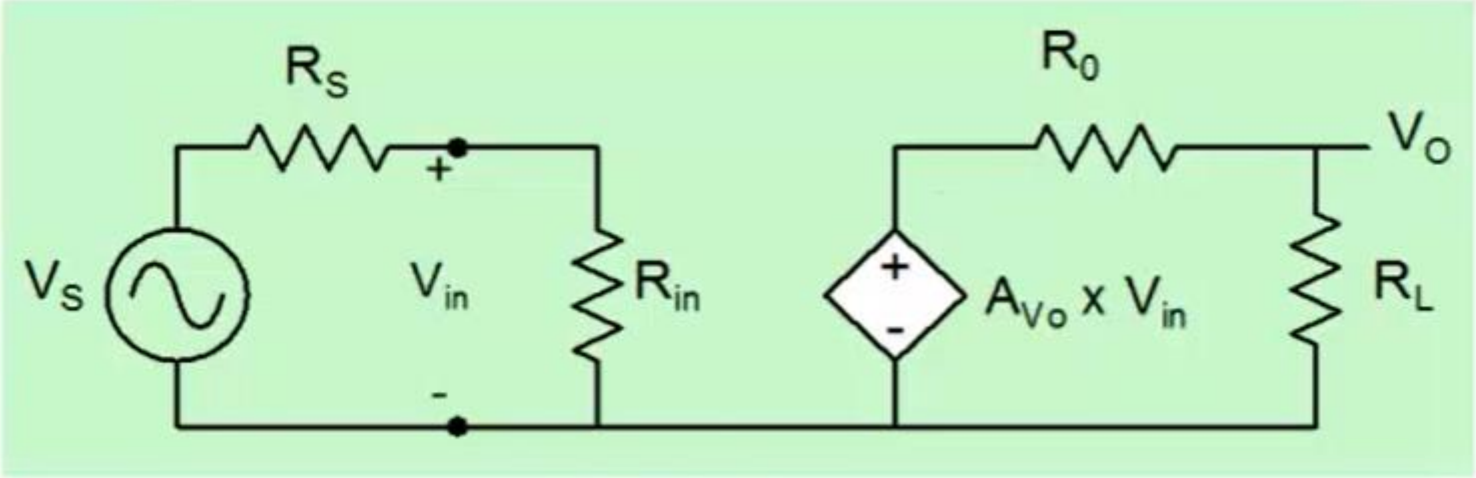
Input Resistance



$$A_{VS} = \frac{V_0}{V_S}$$

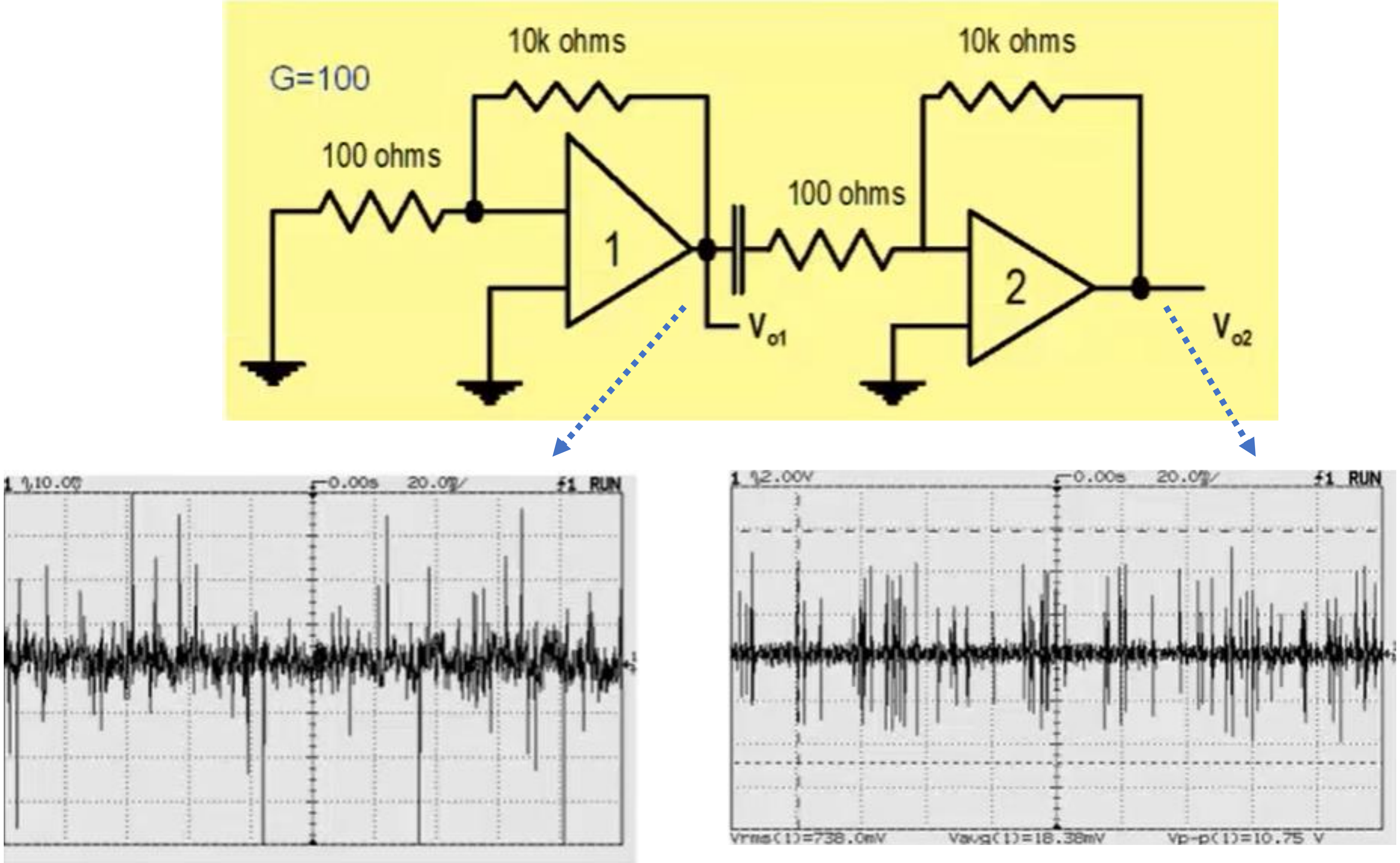


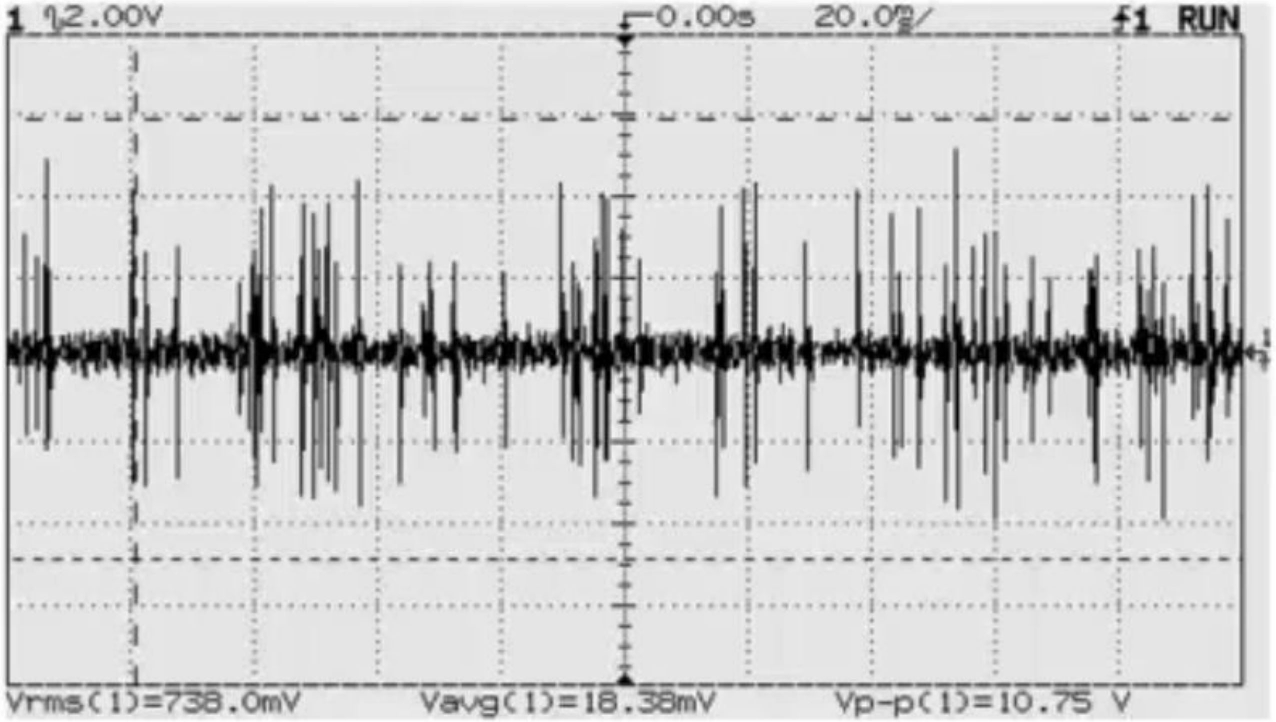




$$A_{VS} = A_{V0} \times \frac{R_{in}}{R_S + R_{in}} \times \frac{R_L}{R_0 + R_L}$$

NOISE


G=100
10k ohms
100 ohms
1
10k ohms
100 ohms
2
V_{o1}
V_{o2}
0.00s
RUN
2.00V
0.00s
RUN
Vrms(1)=738.0mV
Vavg(1)=18.38mV
Vp-p(1)=10.75 V



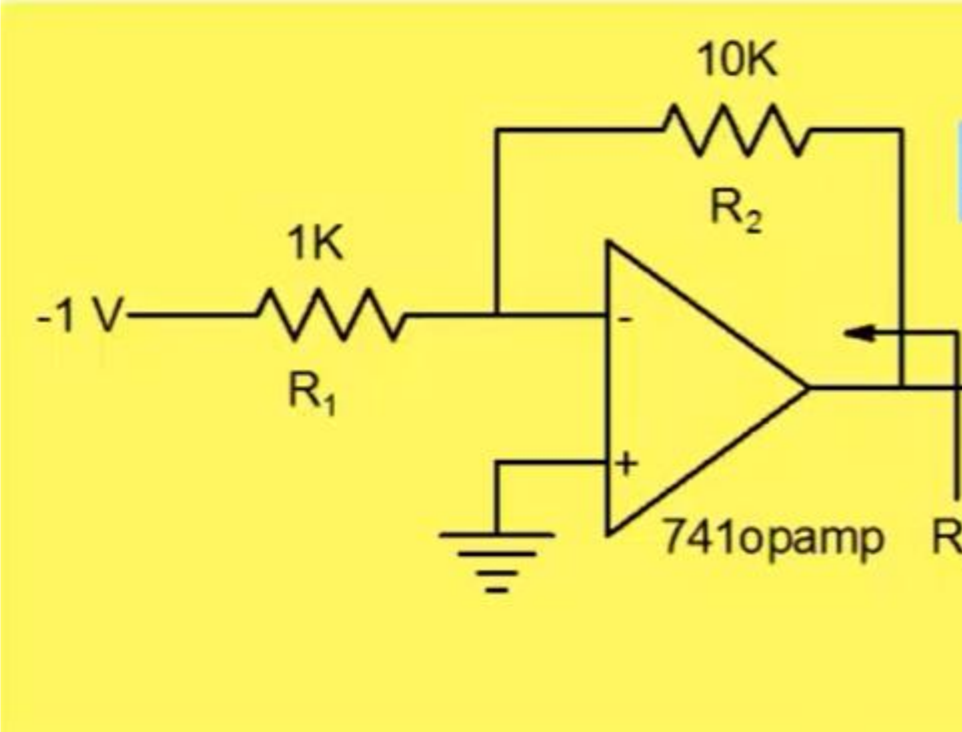
$$F = \frac{\text{Total Output Noise Power}}{\text{Output Noise due to Input Noise Only}}$$

F=1.26

Noise Figure : $NF = 10 \times \mathrm{Log}_{10}(F)$

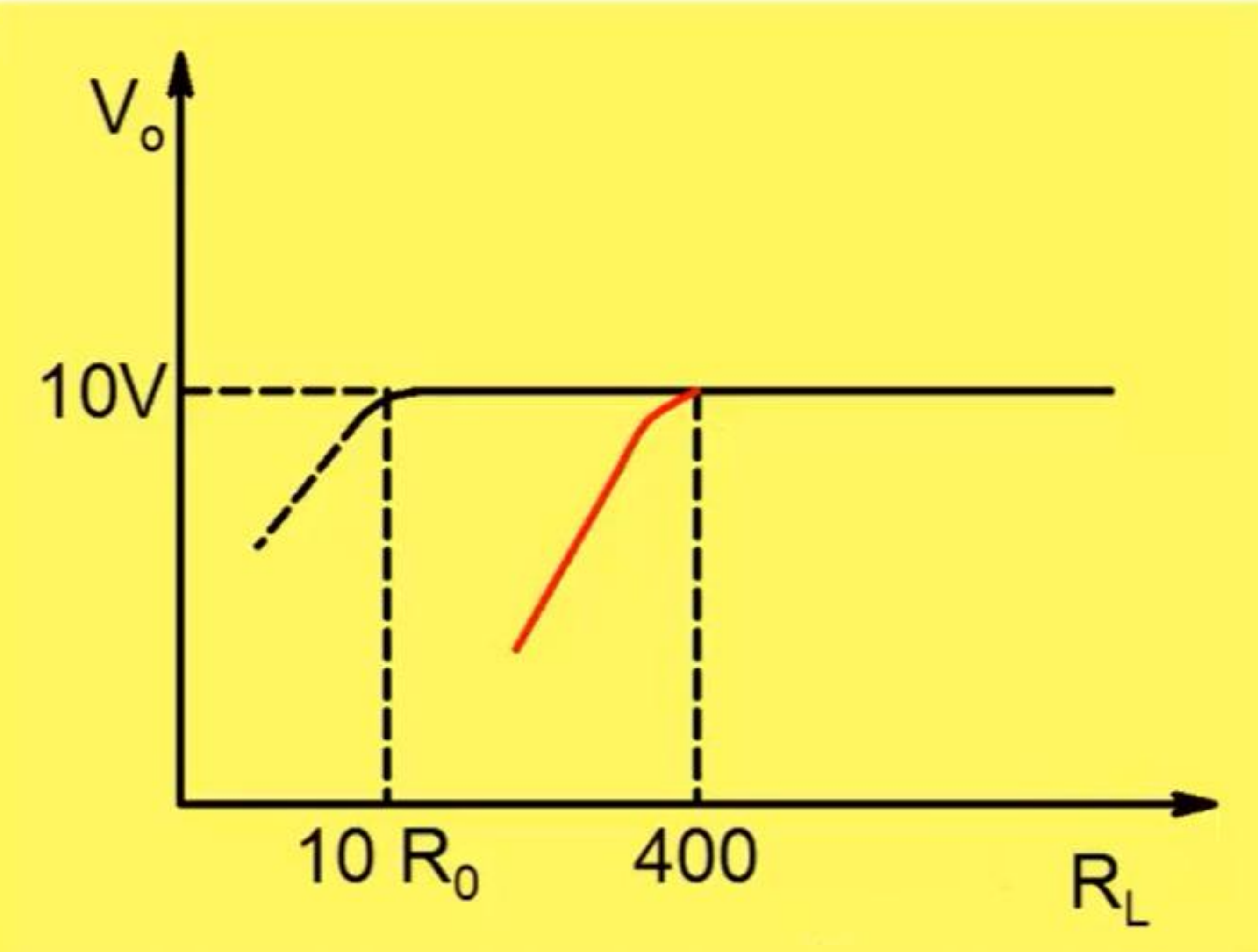
NF=1 dB

Maximum Current Driving Capability



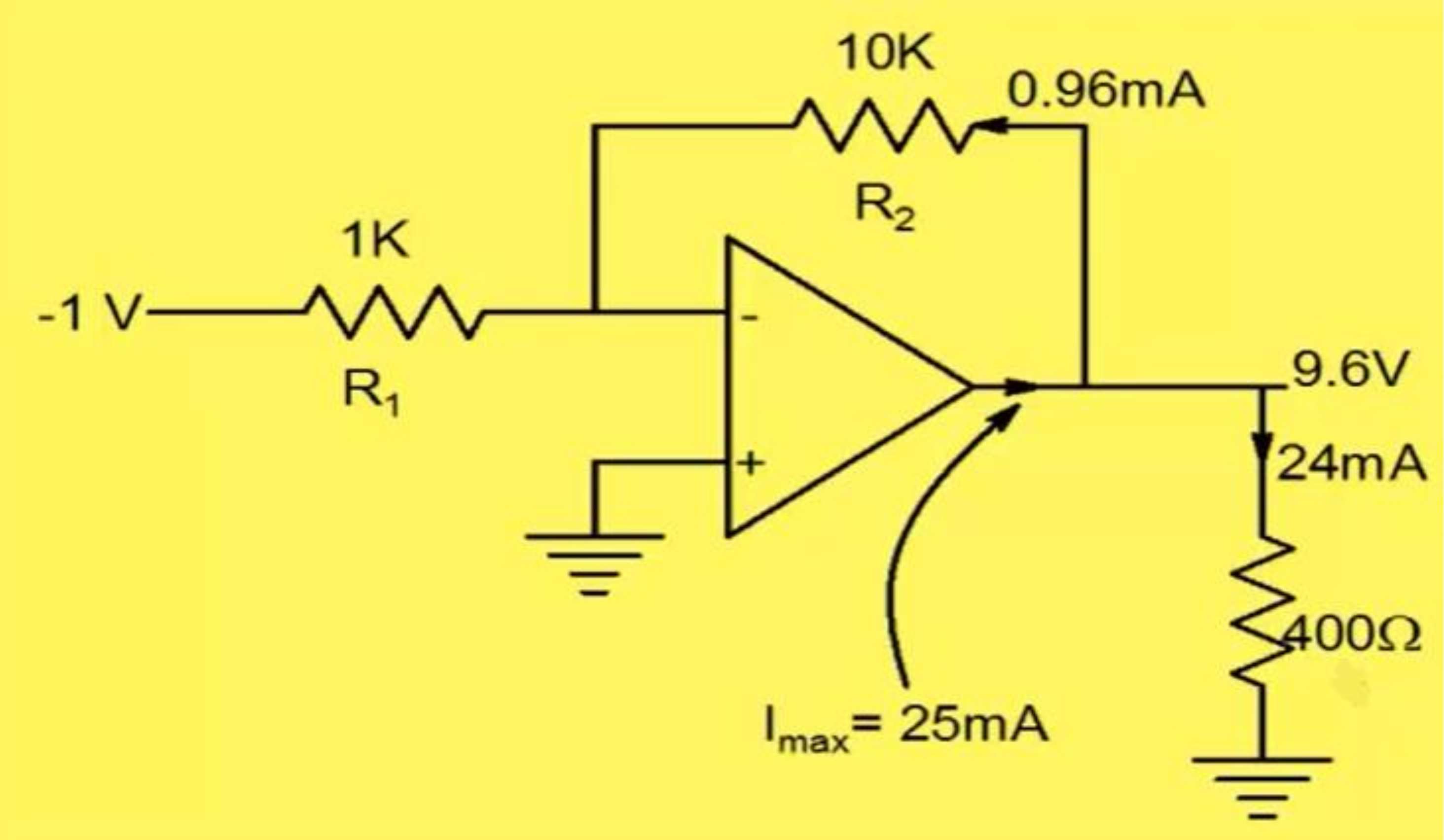
$$R_O < 1\Omega$$

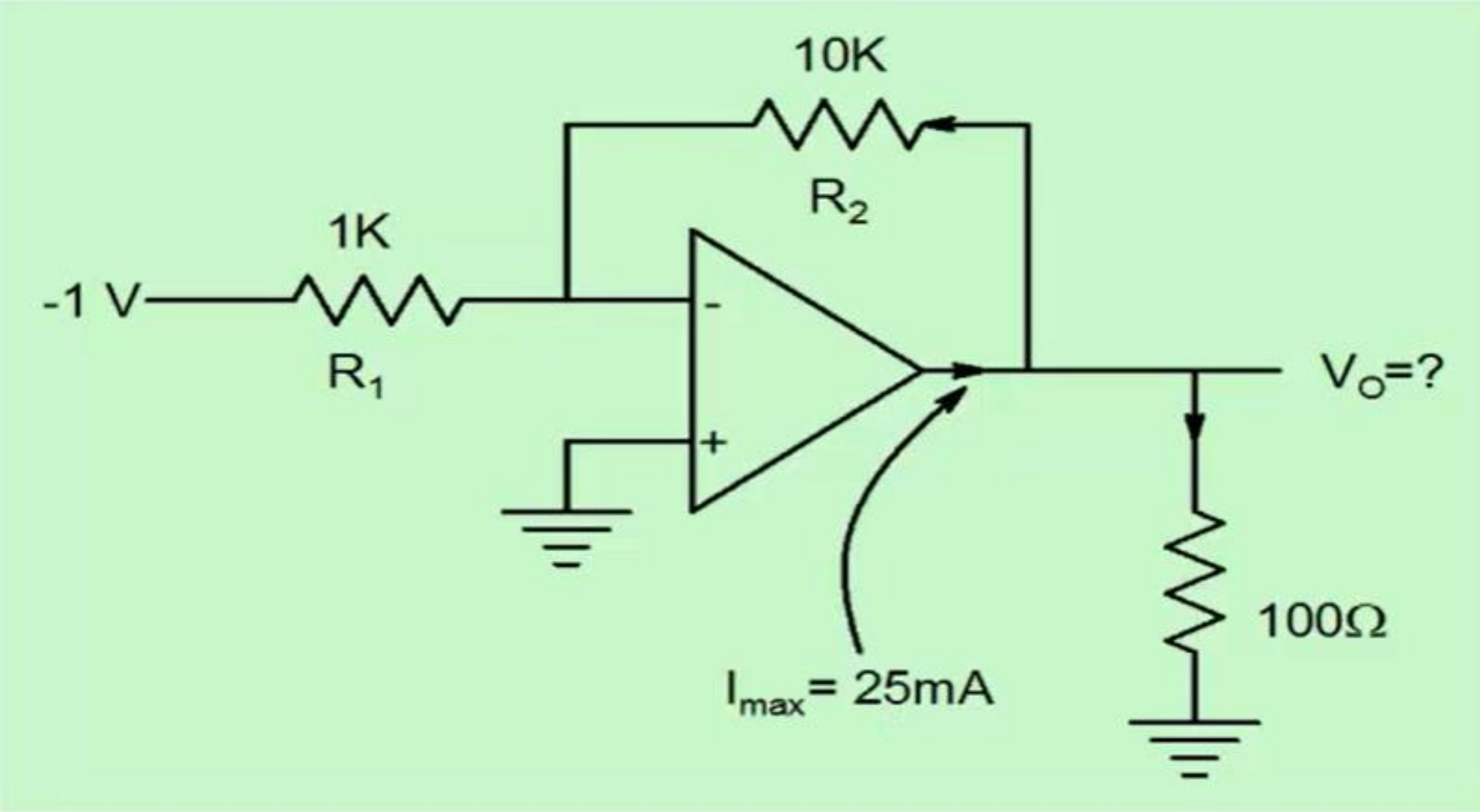
$$V_0 = -\frac{R_2}{R_1} V_{in} = 10V$$



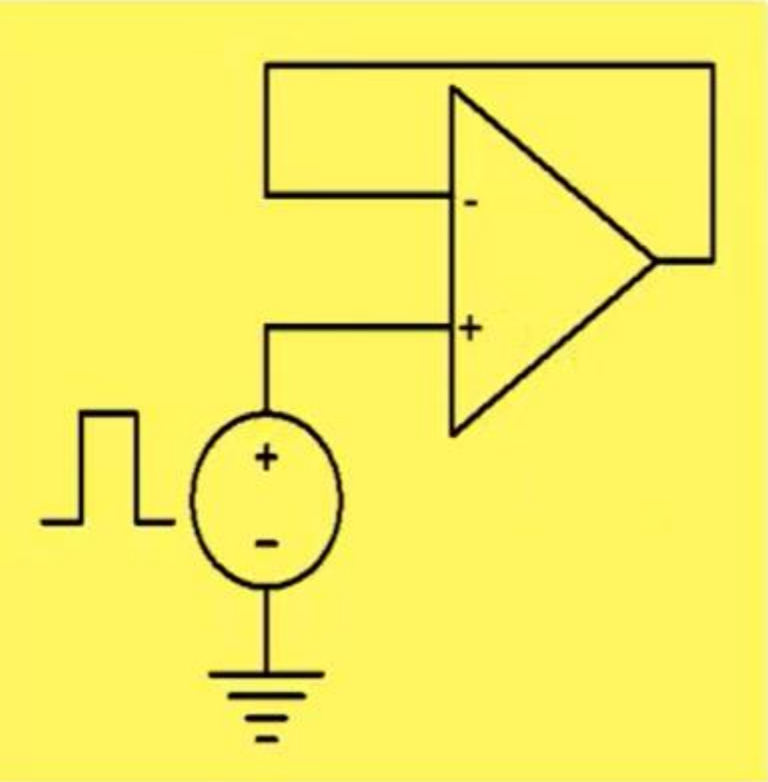
V_o
10V
10 R_0
400
R_L

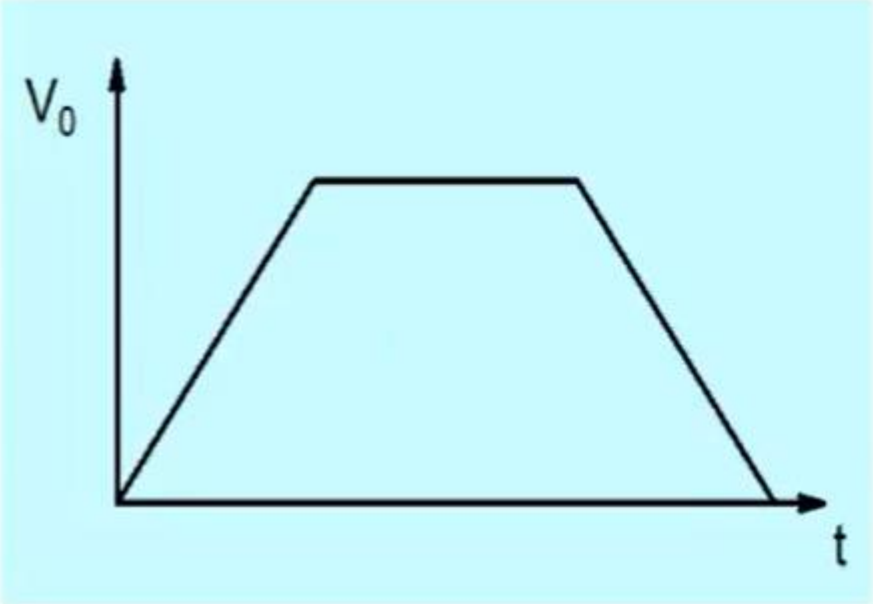
Opamp has maximum current drive capability of 25mA




10K
R2
1K
-1 V
R1
-
+
Vo=?
100Ω
Imax= 25mA

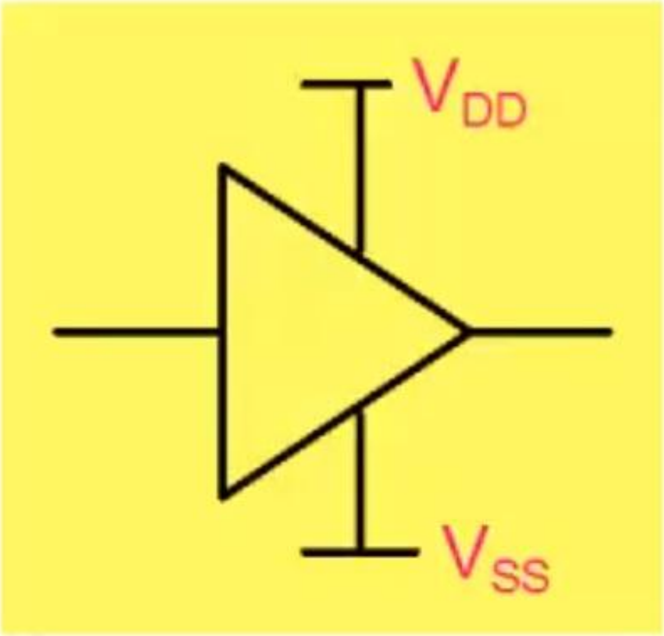
Slew rate





$$\text{Slew rate} = \left.\frac{dv_0}{dt}\right|_{\max}$$

Other specifications



$$\eta = \frac{P_L}{P_{\sup ply}}$$

Power supply rejection ratio, common mode rejection ratio.....

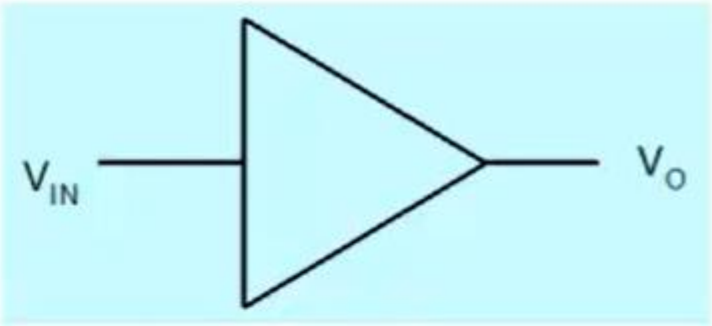
Summary

$$v_o = A_v(f, V_{in}, R_L, R_S, T) \times v_{in} + \tilde{e}_N$$

A_{Vo} $\quad BW \quad THD \quad R_O \quad R_{IN} \quad NF$

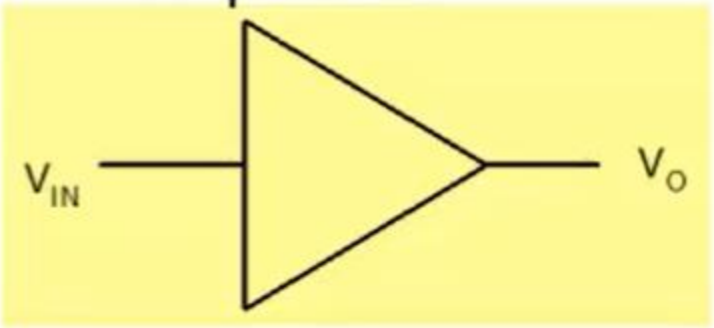
Create situations where one amplifier is better than the other

Amplifier-1



$$A_{V0} = 10^2$$
$$R_{in} = 1k\Omega$$
$$R_o = 1k\Omega$$

Amplifier-2



$$A_{V0} = 10^1$$
$$R_{in} = 100k\Omega$$
$$R_o = 1k\Omega$$

Can the following amplifier be useful?
Can it be called an amplifier?

$$A_{V0} = 1; R_{in} = 10k\Omega; R_o = 10\Omega$$

Semiconductor Basics: A quick review

Semiconductor Basics: A quick review

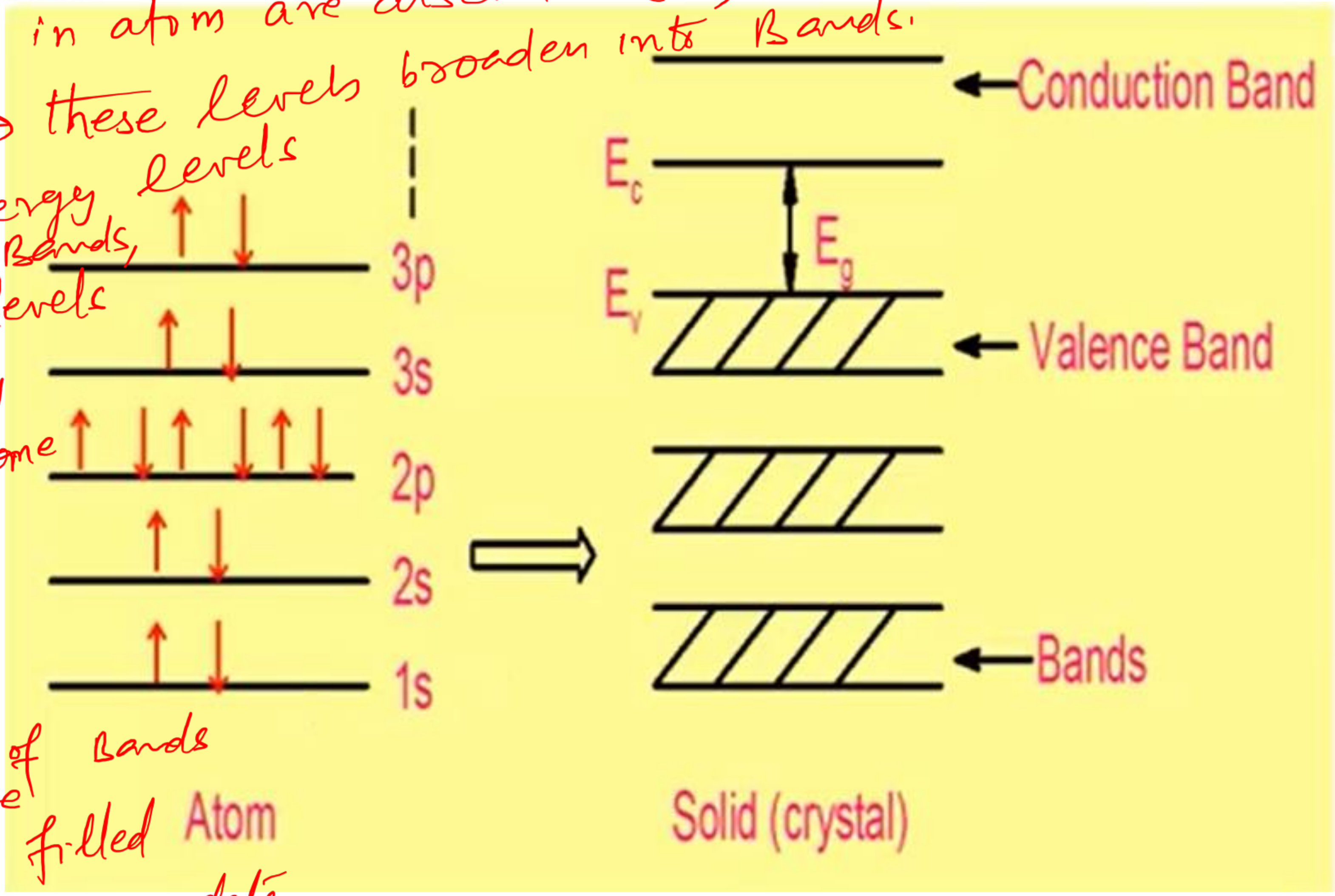
Energy Bands

① Energy levels in atom are discrete (1s, 2s, ...)

② In crystal → these levels broaden into Bands.

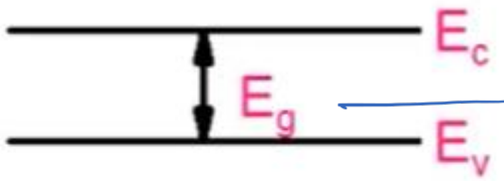
③ As these energy levels expands into Bands, These energy levels also mixed, therefore band structure become complicated.

④ Examine the Band e.g, Si have series of Bands
→ Some are completely filled
i.e, can't accomodate more e^-



A completely full band of electrons cannot contribute to current conduction !

⑤ But in semiconductor,
→ some Bands are fully filled
→ Some are empty
• Nature of semiconductor at 0°C
• Bands which are of interest in semiconductor are top most
• Valence Band
• Conduction Band.

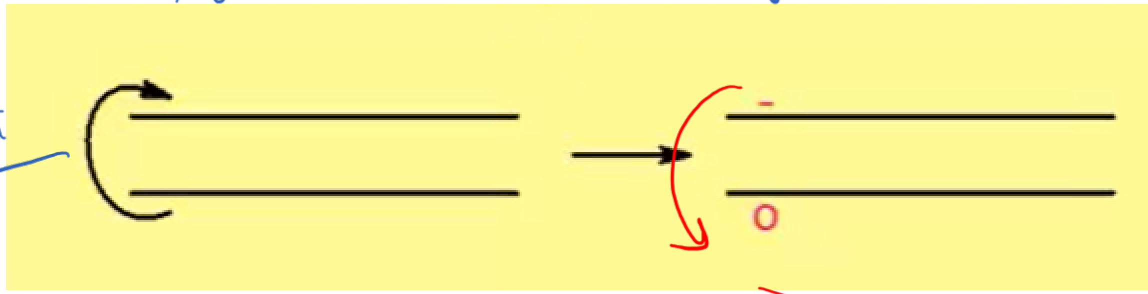


→ Band gap

E_g=1.12 eV for Silicon at 0°K (≈ −273 °C)

① at room temperature
• Some of $\bar{e}$ in VB, gain enough energy from thermal field → They are able to pass E_g

Generation of Carrier



② Now, $\bar{e}$ are in CB, when applied $\vec{E}$, they may move and conduct current.

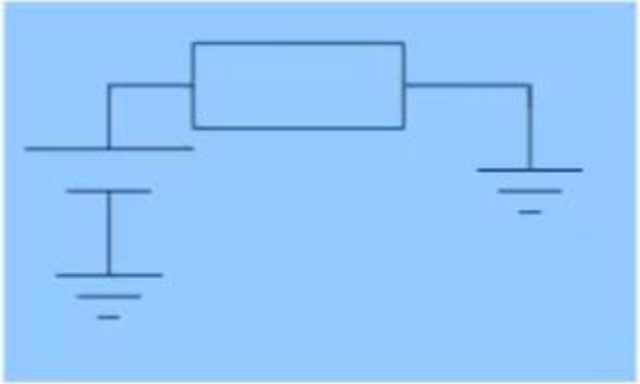
Recombination

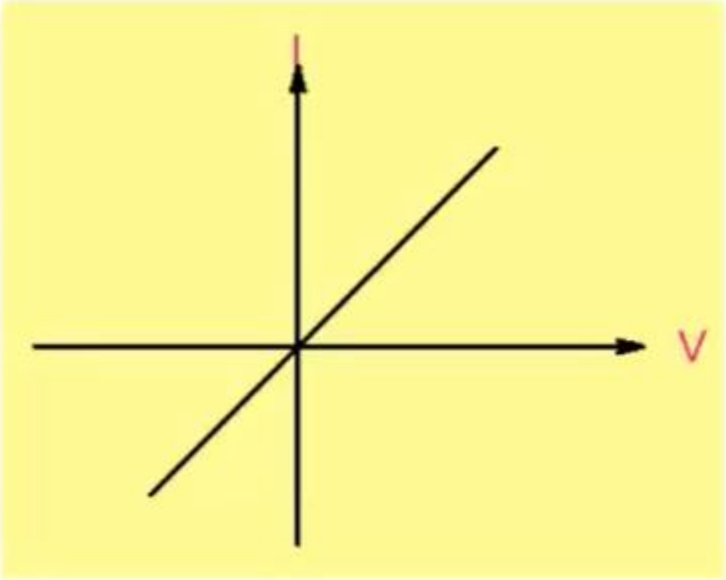
$$n = p = n_i$$

$$n_i = 1.45 \times 10^{10}\ \text{cm}^{-3}\ (T = 300K)$$

③ In equilibrium: Generation and Recombination → Two processes balance each other.

$$n_i \propto \exp\left(-\frac{E_g}{2kT}\right)$$





$$J = \sigma E$$

$$I = \frac{V}{R} \; ; \; R = \frac{1}{\sigma} \times \frac{L}{W}$$

conductivity

amount of $\bar{e}$ or h^+

$$\sigma_i = q(\mu_n + \mu_p)n_i$$

$\bar{e}$ mobility

$$\sigma_i \approx 4 \times 10^{-6}\ \Omega^{-1}\ cm^{-1}$$

$$\rho_i \approx 2 \times 10^{5}\ \Omega\ cm$$

Pure semiconductor is pretty Resistive

→ current will be small

Doping

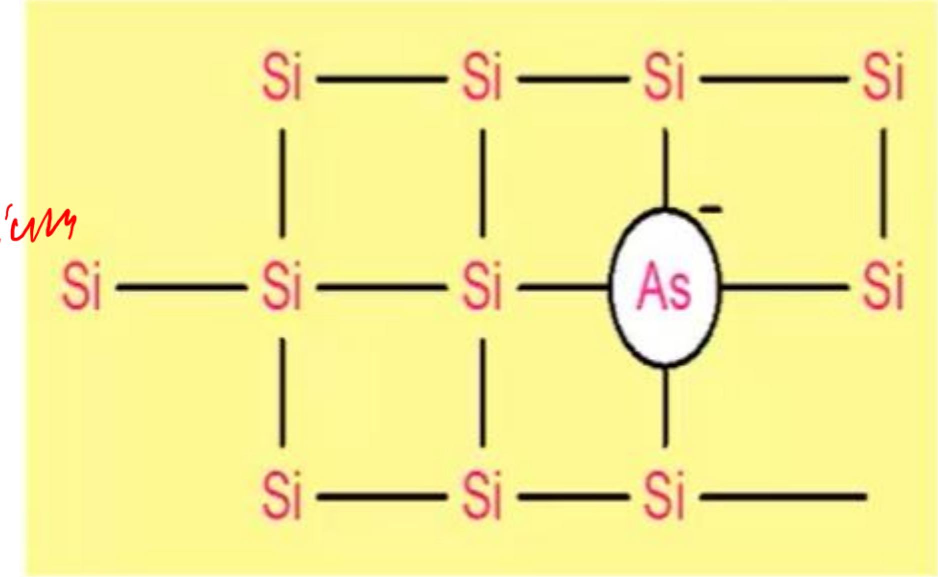
N-Type Semiconductor

- P, As, Sb, Bi → pentavalent (V group) (Donor)

① Extra $\bar{e}$ is loosely bound and it goes to CB.

② So, when we add (e.g., As); it adds extra $\bar{e}$ in CB.

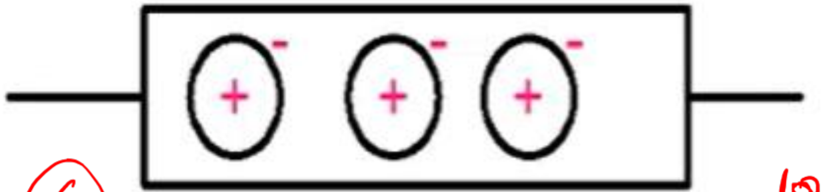
⑤ In equilibrium $n \cdot p = n_i^2$



$N_D \longrightarrow N_D^+ + e^-$ → creates $\bar{e}$ in CB

③ In this process → h^+ are not created.

- Source of $\bar{e}$ is Donor atom, it gets ionized become positively charged and creates an $\bar{e}$ in CB.



⑥ why no. of h^+ decreased?

$N_D = 10^{16}\,\text{cm}^{-3}$

$n \approx 10^{16}\,\text{cm}^{-3}$

$p \approx n_i^2/n \approx 2 \times 10^4\,\text{cm}^{-3}$

$n >> p$

④ Normally, donor atom is large, positively charged, it can't move + immobile

Small amount of impurity results in large change in resistivity !

no. of carrier

$\sigma_i = q(\mu_n + \mu_p) n_i$

- σ depend on n_i

now $\to 10^{16}$

No. of Silicon atoms / volume = $5 \times 10^{22}\ cm^{-3}$

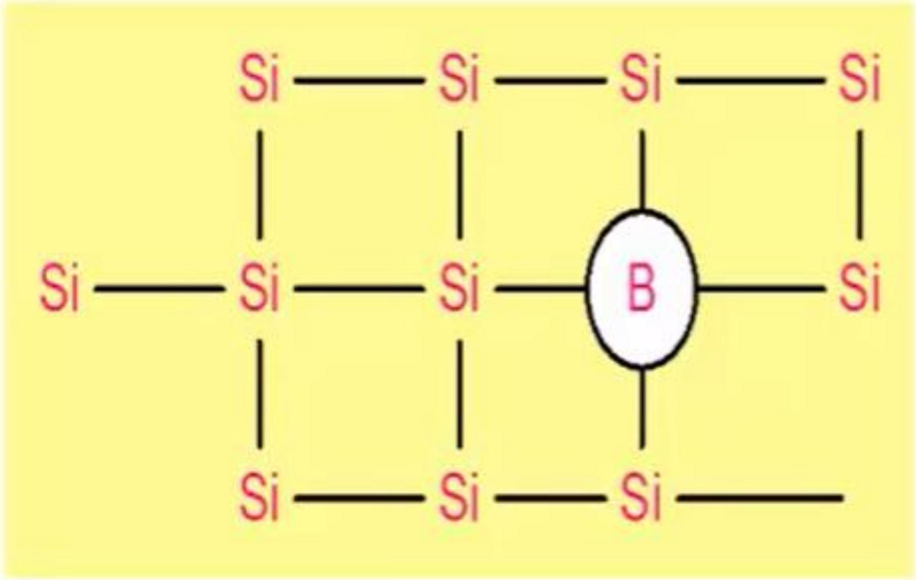
$$\frac{10^{16}}{5 * 10^{22}} = 2 \times 10^{-7} \quad (0.2PPM)$$

$$\rho : \quad 2 \times 10^{5}\ \Omega\, cm \xrightarrow{N_D = 10^{16}\ cm^{-3}} 1.5\ \Omega\, cm$$

unique property in semiconductor

P-Type Semiconductor

B, Al, Ga → Trivalent impurity



$$N_A + e^- \longrightarrow N_A^- + h^+$$

→ Creates h^+ in VB

$N_A = 10^{16}\ cm^{-3}$

$p \approx 10^{16}\ cm^{-3}$

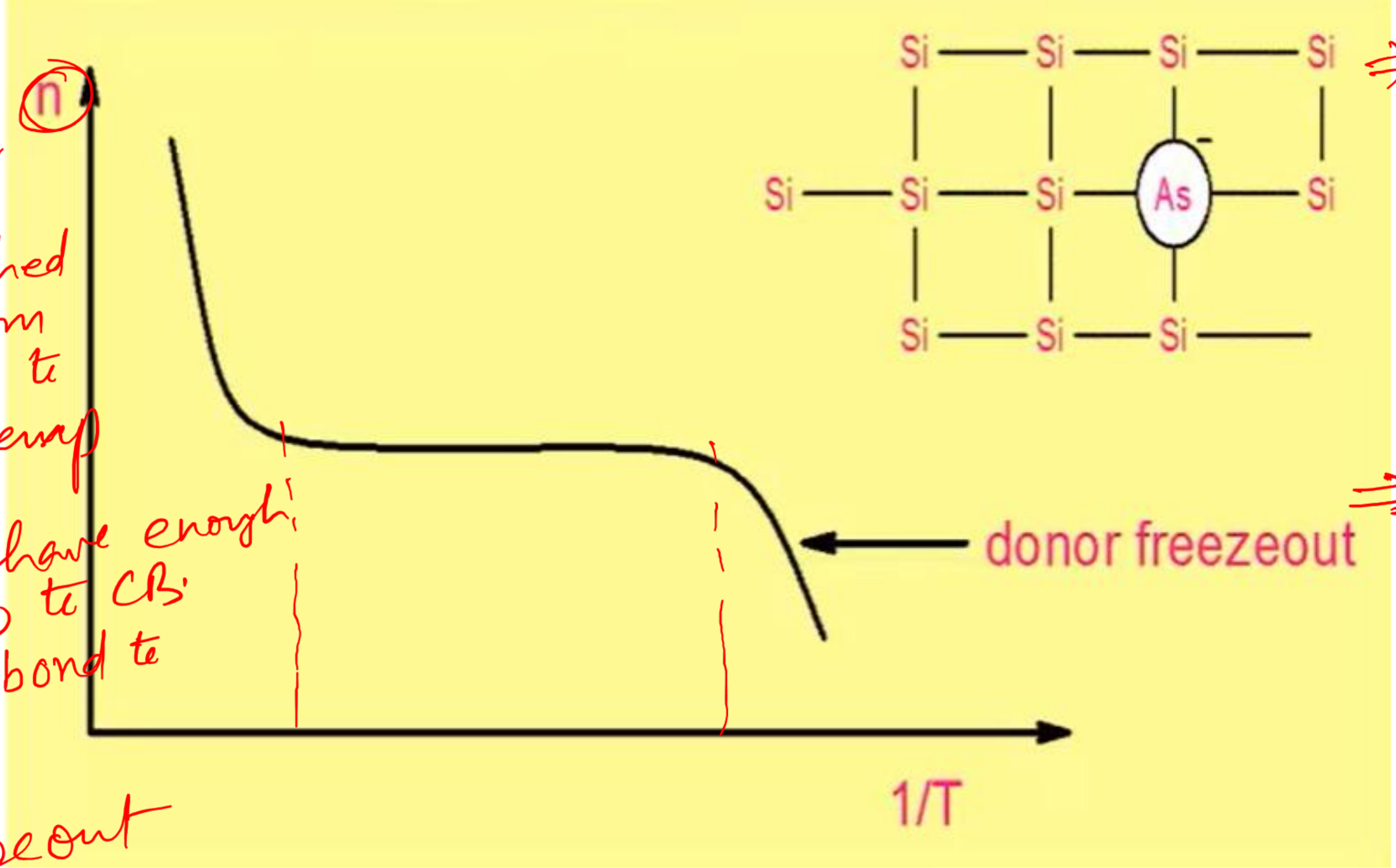
$n \approx n_i^2 / p = 2 \times 10^4\ cm^{-3}$

$p >> n$

Number of carriers is dependent on temperature !

① $\bar{e}$ come from the Donor atom (As).
- It is attached to donor atom

② when we go to very low temp
↓
$\bar{e}$ doesn't have enough energy to go to CB.
- It stays bond to "As" atom
↓
Donor freezeout

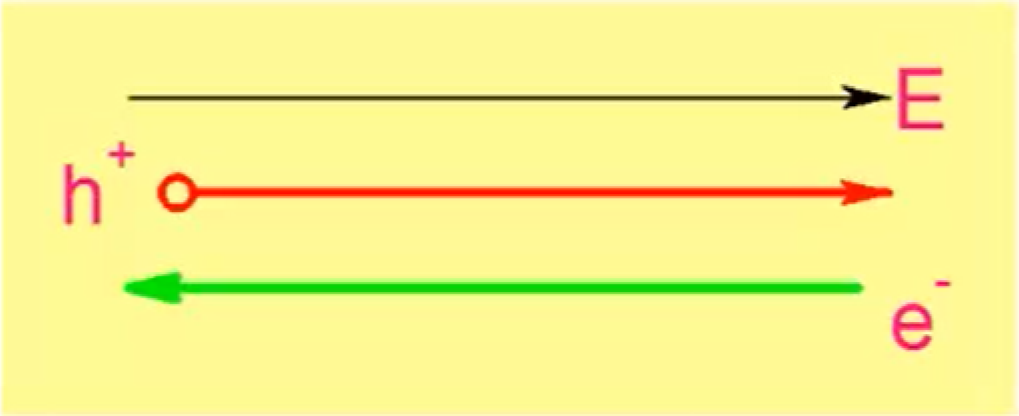


③ At High temp:
→ $n_i \propto \exp\left(\frac{-E_g}{2kT}\right)$
- what are sources of $\bar{e}$
  - Thermal Energy
  - Doping

⇒ Now Temp ↑ Thermally created $\bar{e}$ and h^+ are comparable e.g. 10^{16}
→ material become intrinsic

Current flow

Drift due to Electric field



Current Density

$$J_n = qn\mu_n E$$

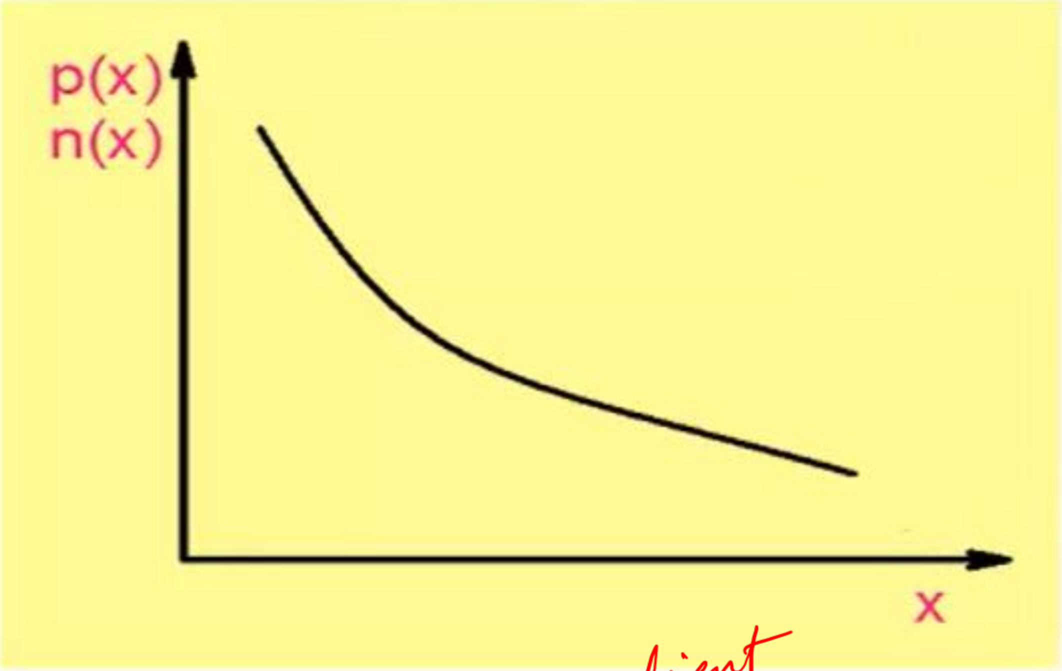
$$J_p = qp\mu_p E$$

$$J = J_n + J_p$$

Current flow

[Even when no electric field]

Diffusion to concentration gradient



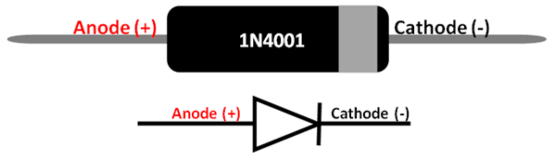
Gradient

$$J_n = qD_n \frac{\partial n}{\partial x}$$

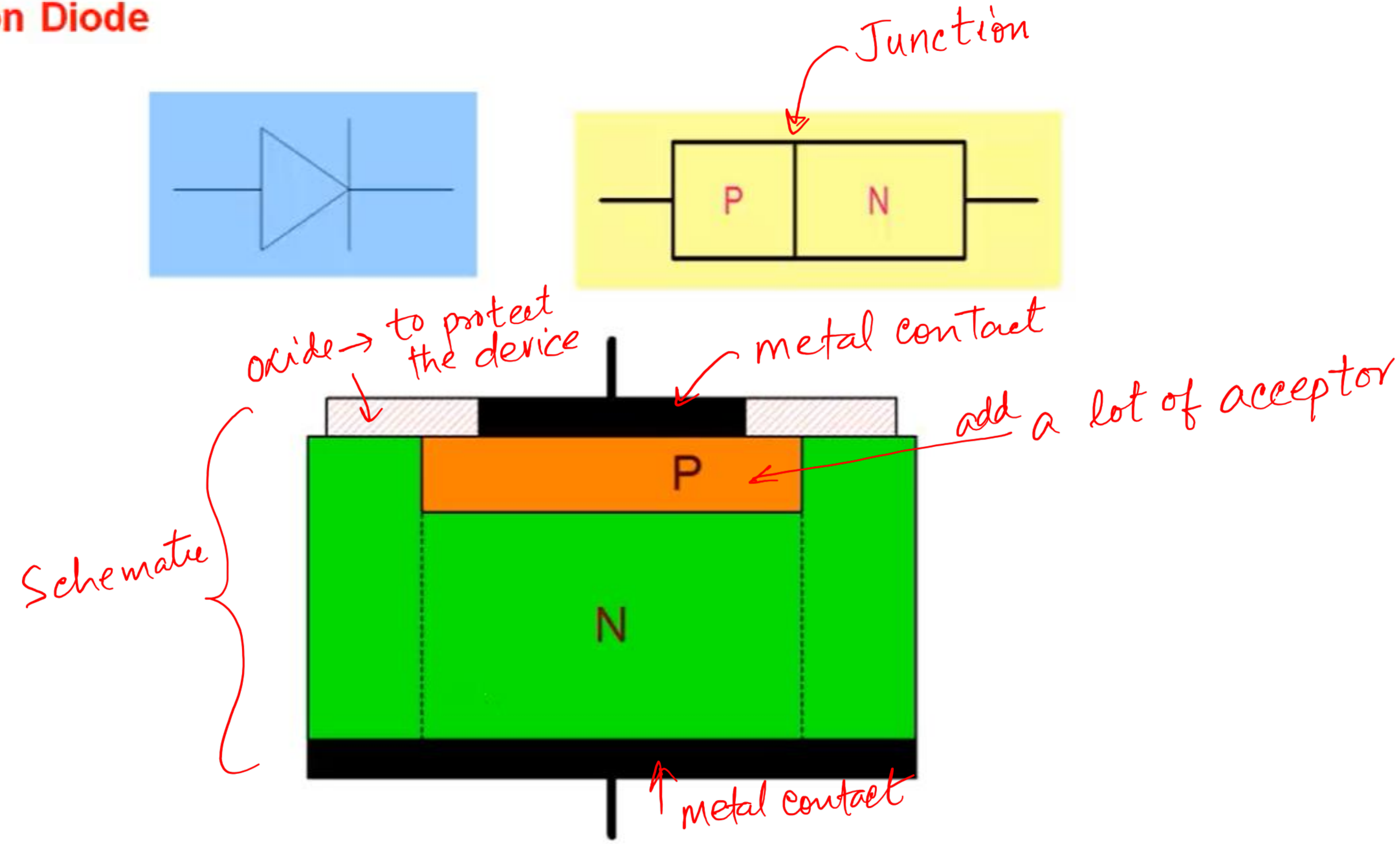
$$J_p = -qD_p \frac{\partial p}{\partial x}$$

Diffusion const

PN Junction Diode

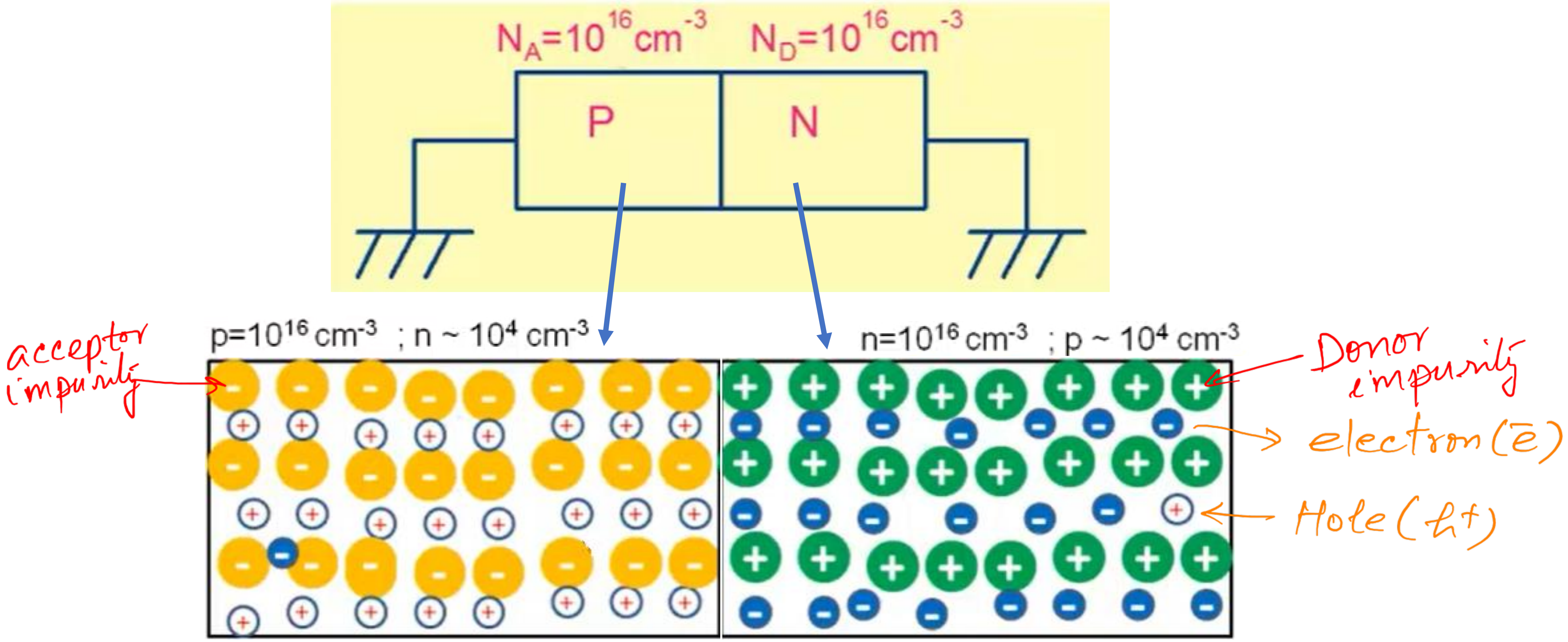


PN Junction Diode



Basic Operation

$n_i^2 = 10^{20}$



Holes will tend to diffuse from p →n and electrons from n→p

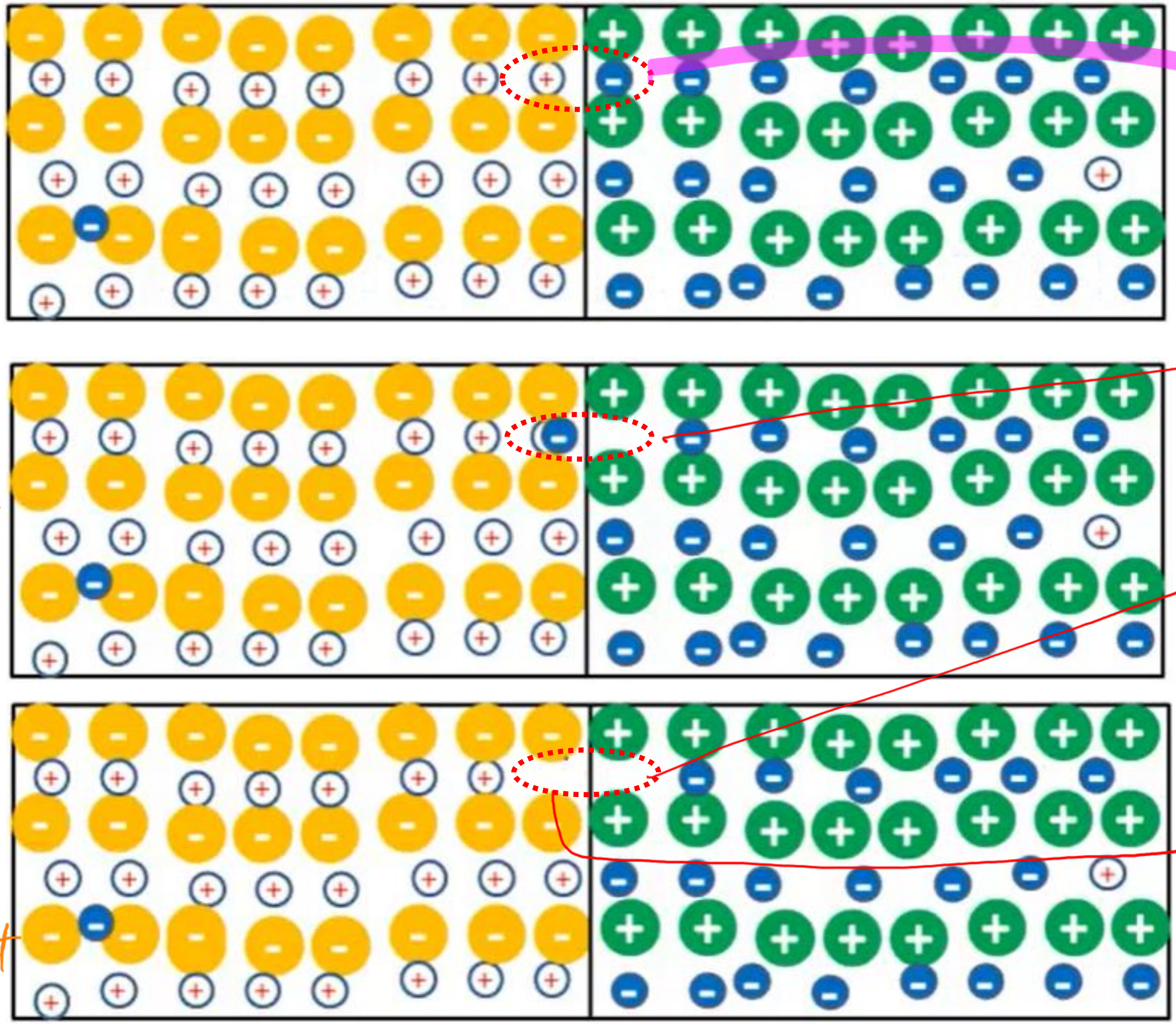
① what are two processes which cause a carrier to move

Drift

Diffusion

② We have huge gradient e.g., a lot of $\bar{e}$ on n-side and a very few on p-type.

③ Similarly, a huge gradient for holes



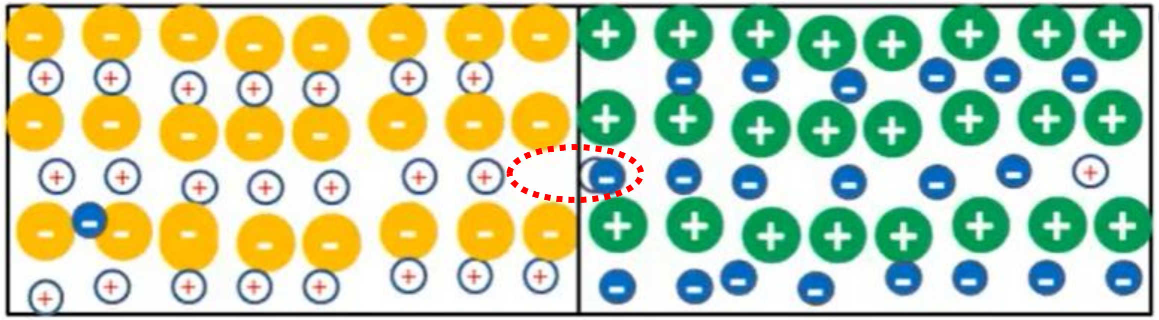
④ when $\bar{e}$ move to p-side → it finds a sea of holes → Holes are empty states → $\bar{e}$ fall into it

⑤ We can see what happed;

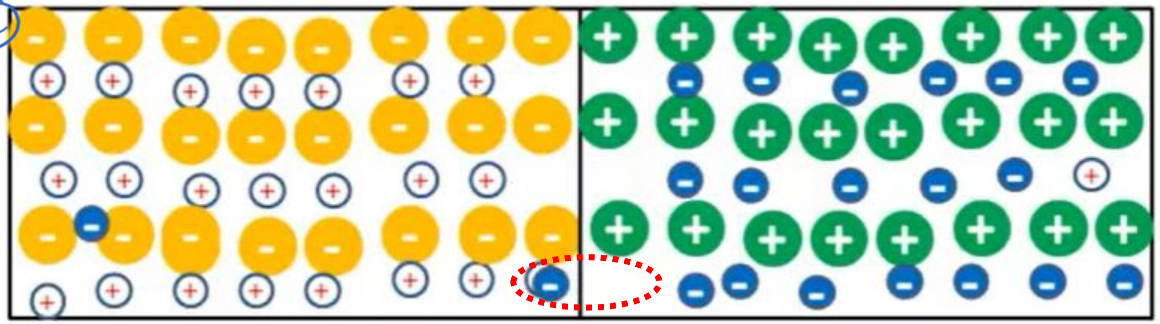
- This $\bar{e}$ was covering this positive charge i.e., neutra-lizing it.
- Now $\bar{e}$ is not there to neutralize it.
- so, this local region become positive charge on n-side
- on the p-side, the local region become negative charge.

① h^+ also go over to n-side → h^+ finds a lot of $\bar{e}$ → Ready to fall into it.

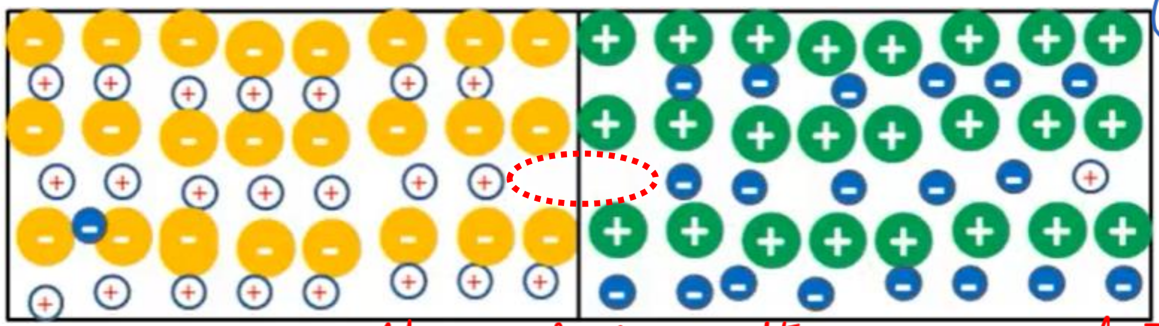
(a)



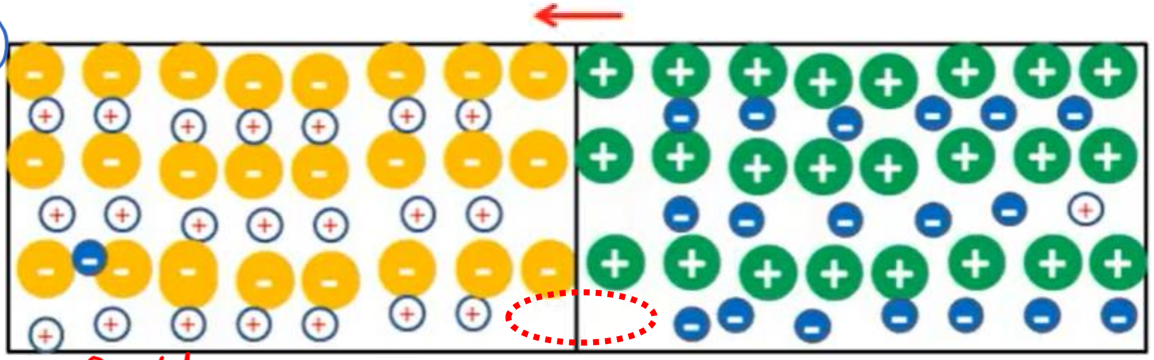
(c)



(b)



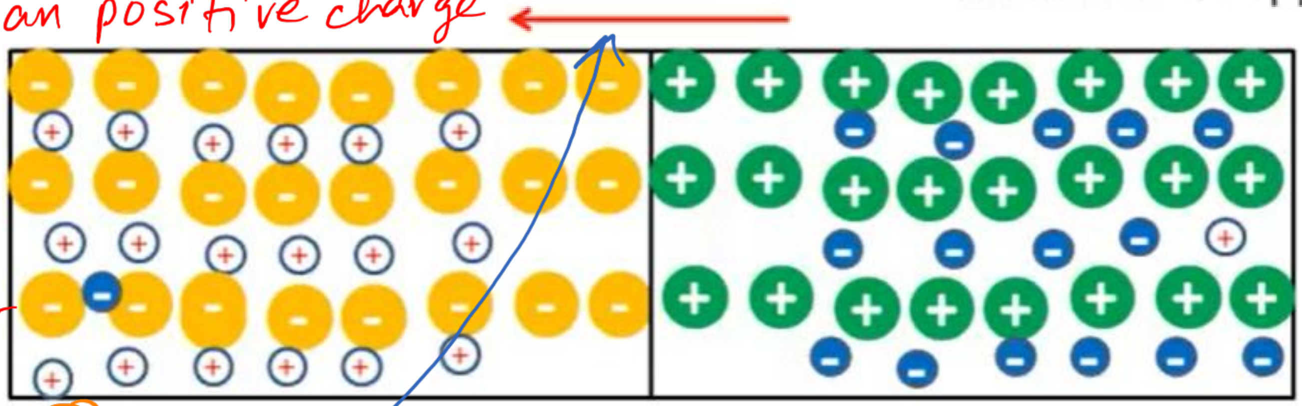
(d)



② Now, as a result of which → there is an electronic field
Because, we got an positive charge on n-side and negative charge on p-side.

$\vec{E}$ ⇒ positive to negative direction

Electric field opposes flow of carriers



③ what this field will do?
- This field will prevent moving h^+ (note the direction of $\vec{E}$)
- This field will oppose the movement of the h^+
- Now h^+ and $\bar{e}$ will find a little difficulty in movement.
- still, our Diffussion gradient is very high.
- So, the process will continue

④ Now, what happens? Field become stronger.

⑤ Eventually equilibrium is reached and there is no net flow of carriers across the junction

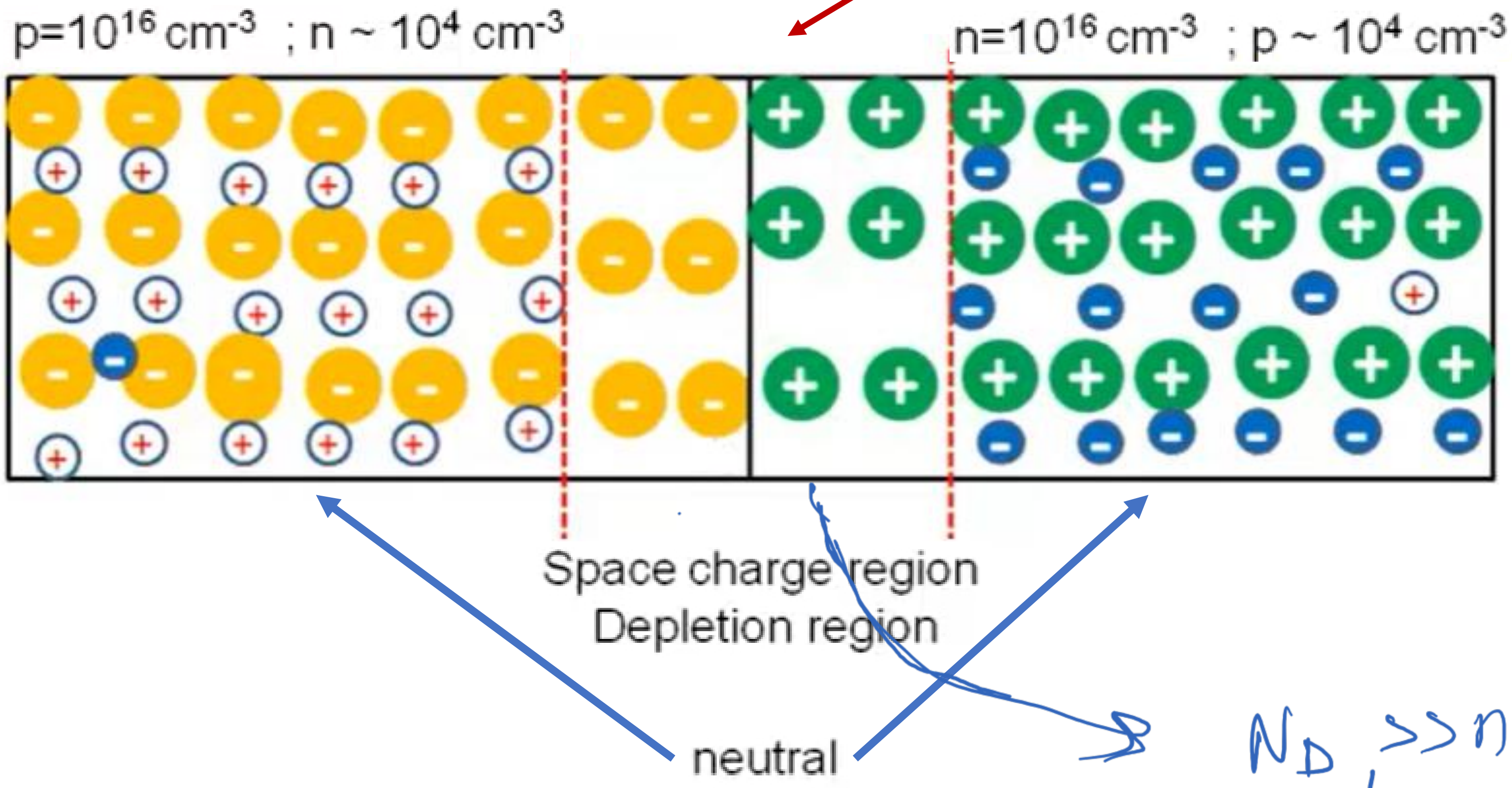
PN Junction Under Equilibrium

① when we reached equilibrium → what will be find?

• a region close to junction has lost carrier

Note: It is not like, this region has completely devoid of carrier → But has lost some carriers

Although small, depletion region has some carriers

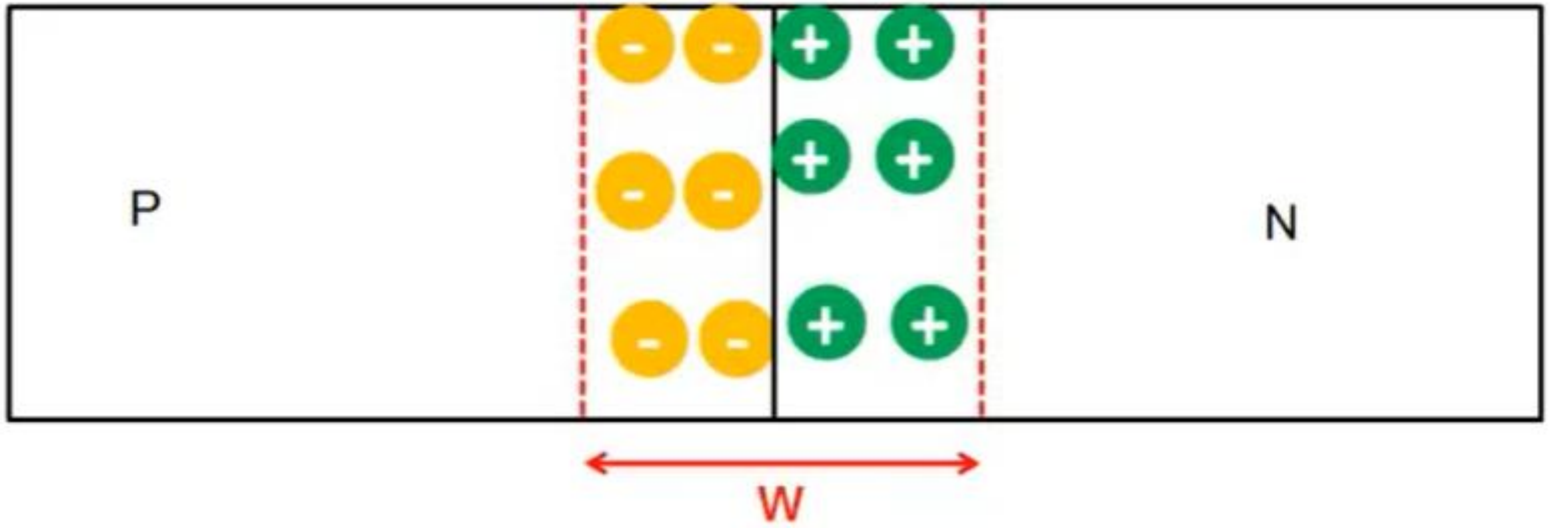


$N_D >> n$

• That is why we call this region as positive charge.

• $\bar{e}$ are not able to neutralize the donors

Depletion region



$$W = \sqrt{\frac{2\varepsilon_s}{q} \times \left(\frac{1}{N_A} + \frac{1}{N_D}\right) \times V_{bi}}$$

$$\varepsilon_s = 11.7 \times 8.85 \times 10^{-14}\, F/cm;\ q = 1.6 \times 10^{-19}\, C$$

$N_A = N_D = 10^{16}\, cm^{-3}$, $T = 300°K$

$V_{bi} = 0.86\ V$

$$W = 4300\, A^{o}$$

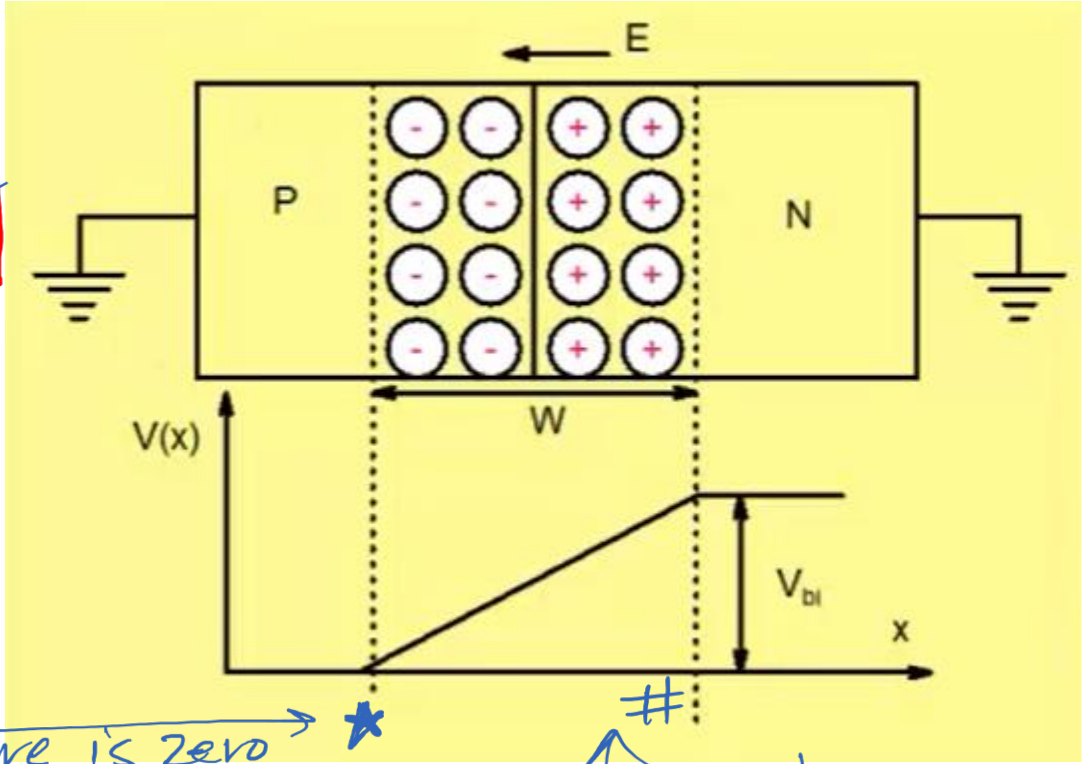
order is like $2000 A^{o}$
" $3000 A^{o}$, etc

Built-in Potential V_{bi}

① once equilibrium is reached → we have depletion region

② In depletion region → we have $\vec{E}$ (look at the direction)

• ignore the external ground for the moment



④ From eqn Ⓐ

• V_{bi} depends on doping but not very sensitive to doping

↳ log of doping

③

• If we assume that potential there is zero → ★

• what will be potential Here → #

↳ something higher

$$V_{bi} = \frac{kT}{q}\ln\left[\frac{N_A N_D}{n_i^2}\right] \quad \text{Ⓐ}$$

$N_A = N_D = 10^{16}\ cm^{-3}$, $T = 300°K$

$V_{bi} = 0.86\ V$

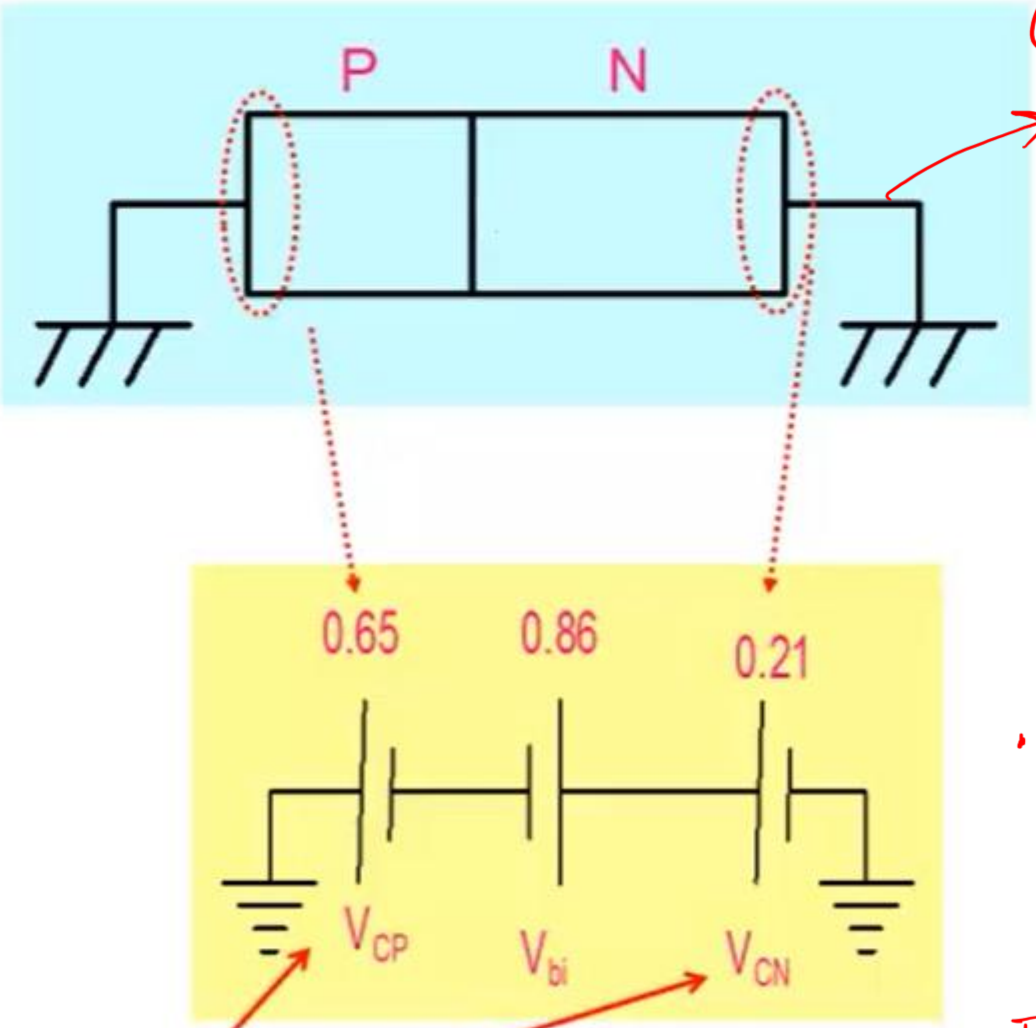
Anytime you put two different materials into contact, a potential develops between them

Notice: N-side is at higher potential compared to p-side → we call this Built-in potential.

Note: It is not something we have applied externally. It is part of our PN-junction.

• Let's look at broader picture

Built-in Potential



Contact potential

① copper wire

• We have a contact potential between n-type and a copper wire

• You will not be able to measure Built-in potential in Multimeter.

• contacts are transparent to carriers

ohmic

• The Behaviour of pn-junction is determined by pn-jn contact.